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An Analysis of Discrete Image Tokenizers under Perturbations and Distribution Shift

Tesi di Laurea Magistrale in
Mathematical Engineering - Ingegneria Matematica (MST –
Statistical Learning)

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Abstract

Discrete image tokenizers convert pixels into token sequences for autoregressive models and reconstruct images from those tokens. This thesis studies how two pretrained tokenizers, VQGAN and LlamaGen VQ-16, respond to perturbations and distribution shift. The analysis combines reconstruction metrics with empirical codebook usage, token changes under global noise, encoder response to local image perturbations, decoder response to local token edits, and distribution shift across natural, medical, and document-image datasets.

On ImageNet validation, LlamaGen obtains better reconstruction scores and uses all 16,384 codebook entries at least once in this evaluation. VQGAN uses 972 entries on the same image set, confirming earlier reports of severe underuse for this checkpoint under the recorded preprocessing. At the tested noise levels, most token identities change, while a saved-image comparison finds smaller decoded changes for LlamaGen at each tested level. Noise concentrates code usage without greatly changing raw active-code counts. Local image perturbations change token assignments well beyond the perturbed patch, and direct token edits change pixels outside the corresponding image region. ImageNet-Sketch, BloodMNIST, and RVL-CDIP retain almost all ImageNet-active codes but use them at different frequencies.

Reconstruction quality alone does not capture code use, token stability, or spatial response. These measurements expose distinct properties of the two checkpoints under perturbation and distribution shift.

Keywords: vector quantization, image tokenization, VQGAN, LlamaGen, robustness, autoregressive image generation

Abstract in lingua italiana

I tokenizer discreti convertono immagini in token per modelli autoregressivi e le ricostruiscono dai token. Questa tesi analizza due tokenizer preaddestrati, VQGAN e LlamaGen VQ-16, misurando ricostruzione, utilizzo del codebook, stabilità sotto rumore globale, risposta dell'encoder a perturbazioni locali e del decoder a modifiche dei token, e cambiamenti di distribuzione su immagini naturali, mediche e documentali.

Su ImageNet validation, LlamaGen ottiene metriche di ricostruzione migliori e utilizza tutti i 16.384 elementi del codebook almeno una volta in questa valutazione; VQGAN ne utilizza 972, confermando precedenti osservazioni di sottoutilizzo, con il preprocessing documentato. Ai livelli di rumore testati cambia l'identità della maggior parte dei token, mentre un confronto delle ricostruzioni salvate rileva cambiamenti minori per LlamaGen a ogni livello testato. Il rumore concentra l'uso dei codici senza modificare molto il numero grezzo di codici attivi. Le perturbazioni locali dell'immagine modificano le assegnazioni dei token ben oltre la regione perturbata; le modifiche dirette dei token cambiano i pixel anche fuori dalla regione corrispondente. ImageNet-Sketch, BloodMNIST e RVL-CDIP conservano quasi tutti i codici attivi su ImageNet, ma li utilizzano con frequenze diverse.

La qualità della ricostruzione da sola non descrive l'uso dei codici, la stabilità dei token o la risposta spaziale. Queste misure distinguono il comportamento dei due checkpoint sotto perturbazione e variazioni della distribuzione.

Parole chiave: quantizzazione vettoriale, tokenizzazione di immagini, VQGAN, LlamaGen, robustezza, generazione autoregressiva di immagini

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1 | Introduction

Large language models have shown that a single next-token prediction objective can support increasingly capable systems as models and datasets scale. This progress has motivated extending the same machinery beyond text. Images, however, are continuous arrays of pixel values rather than sequences of discrete symbols. One way to bridge this difference is to first encode an image as a short sequence of visual tokens. A language-model-style autoregressive network can then read or generate those tokens, and a decoder can convert the generated sequence back into an image.

Vector-quantized image tokenizers provide this interface. An encoder maps an image to a grid of continuous feature vectors; each vector is replaced by the index of a nearby entry in a learned codebook; and a decoder reconstructs the image from the resulting token grid. This reduces a 256×256 image to a much shorter sequence, making autoregressive image modeling practical. The tokenizer is a compression stage that also determines which visual distinctions are preserved, which images receive similar token sequences, and how perturbations in image or token space propagate through the representation.

Recently, many tokenizers have been proposed using different architectural and training choices. They are commonly compared using reconstruction quality, but similar reconstruction scores can hide very different discrete representations. A tokenizer may use only a small part of its nominal vocabulary, change many token assignments after a small image perturbation, or spread a local intervention across distant spatial positions. Since the downstream autoregressive model learns the token sequences produced by the tokenizer, these properties define the learning problem it receives. Prior studies report reconstruction, generation, and codebook utilization, including underuse by the evaluated VQGAN checkpoint [10, 20, 31]. This thesis adds controlled noise and locality probes and compares code-frequency changes across several input domains for two released checkpoints.

1.1. Problem Statement and Research Questions

We focus on two different tokenizers, VQGAN [10] and LlamaGen VQ-16 [31]. Both produce the same number of tokens per image, which permits direct spatial comparisons. The analysis isolates the tokenizer, including its encoder, vector quantizer, and decoder, from the autoregressive model trained on its output. This study addresses five research questions.

First, how do reconstruction quality and empirical codebook usage differ between the two checkpoints (RQ1)? This establishes the clean baseline used throughout the rest of the study (Chapter 4).

Next, how do global image perturbations affect token assignments, code frequencies, and reconstruction fidelity (RQ2)? Although the encoder-decoder bottleneck compresses the input, it is unclear how much noise it filters and how much reaches the discrete representation and reconstruction. Chapter 5 therefore compares clean and noisy processing paths in both image and token space.

After measuring global effects, we ask whether a local image perturbation remains local in the token grid (RQ3). Chapter 6 measures token changes inside and outside the perturbed image region.

We next ask how far a local token edit changes the decoded image (RQ4). Chapter 7 measures how a local token edit changes the reconstructed image inside and outside the corresponding region.

Finally, how do code use and reconstruction metrics respond to a change in the input distribution (RQ5)? Chapter 8 compares reconstruction quality, codebook usage, and the spatial allocation of tokens across input domains.

Figure 1.1 summarizes the five stages of the study. For the global-noise experiment, the original and noisy images are processed in parallel; reconstruction metrics compare images, while token flips compare the two token grids. The local experiments separately measure the response inside and outside the intervened region.

1.2. Thesis Outline

The thesis is structured as follows. Chapter 2 introduces autoregressive image modeling and vector-quantized tokenizers. Chapter 3 defines the evaluated models, datasets, and metrics. Chapters 4–8 address the five research questions in sequence. Chapter 9

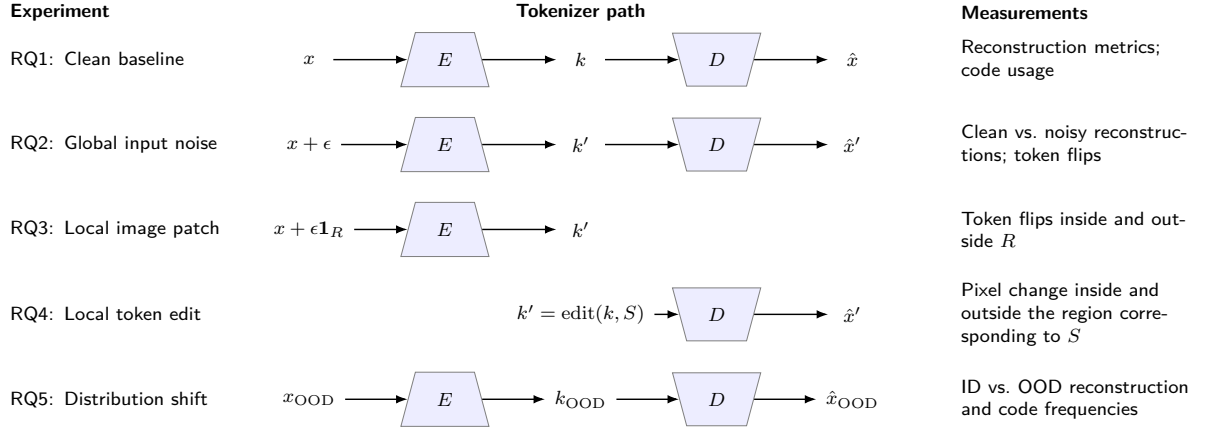


Figure 1.1: Overview of the five stages of the study. Reconstruction metrics are computed between images, token flips are computed between token grids, and the local experiments measure responses both inside and outside the intervened region.

summarizes the results and discusses the study's limitations and future work.

2 | Background

This chapter introduces autoregressive image modeling and vector quantization, then describes early and recent image tokenizers.

2.1. Autoregressive Models: From Text to Images

An autoregressive model represents a joint distribution by predicting each element from those that precede it. A token is a discrete text unit, such as a word or subword, drawn from a finite vocabulary. Let $s = (s_1, \dots, s_T)$ be a sequence of tokens, where s_i is the token at position i and T is the sequence length. The chain rule gives

$$p(s) = \prod_{i=1}^T p(s_i \mid s_{<i}), \quad s_{<i} = (s_1, \dots, s_{i-1}). \quad (2.1)$$

Bengio et al. use learned conditional probabilities in a neural language model [3]. A text tokenizer converts text into a sequence of tokens drawn from a finite vocabulary, and a language model learns the conditional distribution of the next token. Transformers made this approach effective at scale by using self-attention to model dependencies between tokens [35]. Later work found that language-model loss follows empirical power laws as the amount of data, model size, and training compute increase [17]. Large autoregressive language models then demonstrated broad capabilities under the same training objective [5, 32].

Applying the same principle to images is less direct. An image is a two-dimensional array rather than an ordered sequence, so an autoregressive model must first choose both a prediction unit and an ordering. The most literal choice is to discretize pixel intensities and traverse them in raster order, from left to right and top to bottom. Early works such as PixelRNN and PixelCNN showed that this produces tractable image likelihoods and coherent samples [33]. Image Transformer later applied self-attention to autoregressive pixel generation while restricting attention to local neighborhoods to control its cost [24].

Pixel-level modeling is conceptually simple but computationally demanding. A 256×256

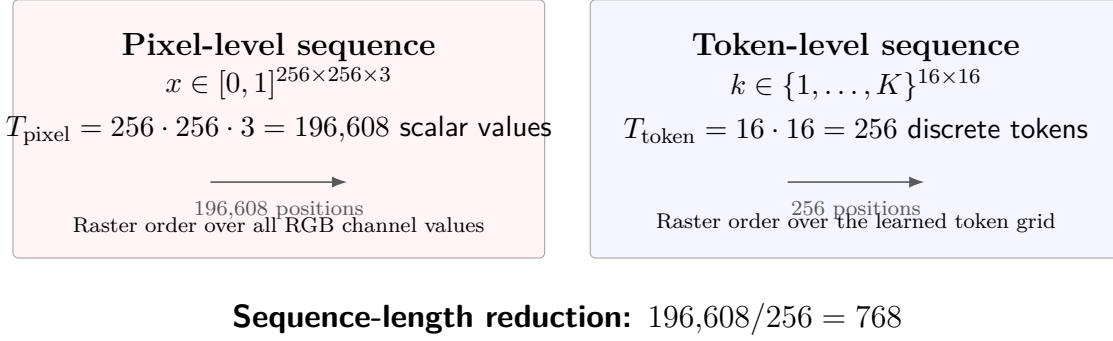
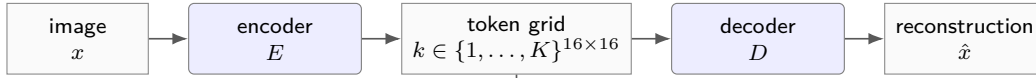


Figure 2.1: Pixel-level flattening produces much longer autoregressive sequences than token-level flattening. A learned tokenizer replaces groups of pixel values with discrete visual symbols.

Stage 1: learn and then freeze the tokenizer



Stage 2: learn an autoregressive prior over tokenizer outputs



Figure 2.2: Two-stage autoregressive image generation. The tokenizer maps an image to a compact token grid; an autoregressive prior models the flattened token sequence; the tokenizer decoder maps sampled tokens back to image space.

RGB image contains 196,608 channel values, and generation must resolve low-level color statistics as well as long-range structure. Moreover, full self-attention on a sequence of length T has $\mathcal{O}(T^2)$ time and memory cost. A learned image tokenizer addresses both problems by mapping the image to a shorter grid of discrete symbols. The token grids studied in this thesis are flattened in raster order. Tokenization reduces the number of prediction units but does not choose a different ordering. Figure 2.1 contrasts the resulting sequence lengths.

Autoregressive image generation therefore has two stages, illustrated in Figure 2.2. A tokenizer first converts images into compact token grids and reconstructs images from them. An autoregressive model is then trained on the resulting token sequences. This second-stage model is also called a *prior*: it models the distribution of token sequences and can sample a sequence for the decoder.

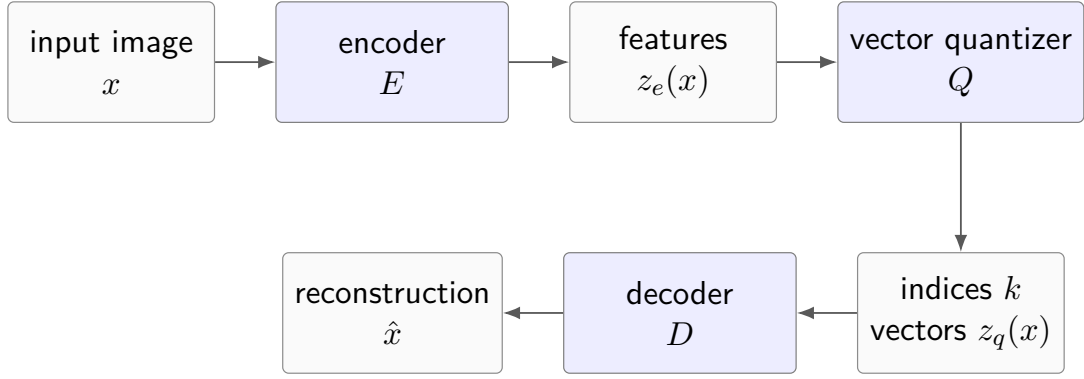


Figure 2.3: VQ-VAE representation path. The encoder produces continuous features, the vector quantizer returns integer tokens and their associated codebook vectors, and the decoder reconstructs the image. Based on van den Oord et al. [34]. *Source: author-created with AI assistance.*

Image tokenizers may use convolutions, attention, or both. Their function is to represent images with discrete tokens and reconstruct images from them. The autoregressive model has a different function: it predicts those tokens in sequence. The tokenizer determines which image information reaches that model, the sequence length, and the vocabulary used for next-token prediction.

2.2. Image Tokens and Vector Quantization

Vector quantization replaces continuous feature vectors with entries from a finite codebook. We introduce this operation using VQ-VAE, which learns discrete latent representations by assigning encoder outputs to a learned codebook [34]. A separate autoregressive model can then model the resulting token sequence. VQ-VAE-2 later showed that hierarchical discrete latents and stronger priors could support high-fidelity large-scale image generation [27].

VQ-VAE contains an encoder, a vector quantizer, and a decoder. Given an input image $x \in \mathbb{R}^{H \times W \times C}$, the encoder produces a continuous feature grid $z_e(x) \in \mathbb{R}^{h \times w \times d}$. The full path is shown in Figure 2.3.

At each spatial position (u, v) , the vector $z_e(x)_{u,v} \in \mathbb{R}^d$ is assigned to its nearest entry in a learned codebook $\mathcal{E} = \{e_1, \dots, e_K\}$, where $e_j \in \mathbb{R}^d$. Quantization produces an integer

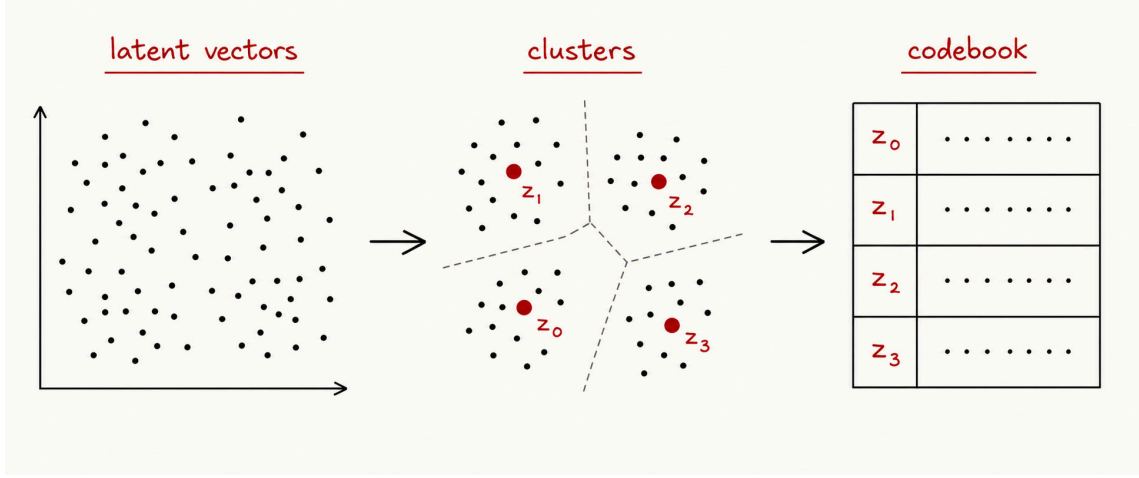


Figure 2.4: Vector quantization as clustering in latent space. Continuous encoder outputs are assigned to their nearest learned codebook entry, and the entry index becomes a discrete token. *Source: author-created with AI assistance.*

token grid $k \in \{1, \dots, K\}^{h \times w}$ and a quantized latent tensor $z_q(x) \in \mathbb{R}^{h \times w \times d}$:

$$z_e(x) = E(x), \quad (2.2)$$

$$k_{u,v} = \arg \min_{j \in \{1, \dots, K\}} \|z_e(x)_{u,v} - e_j\|_2, \quad (2.3)$$

$$z_q(x)_{u,v} = e_{k_{u,v}}, \quad (2.4)$$

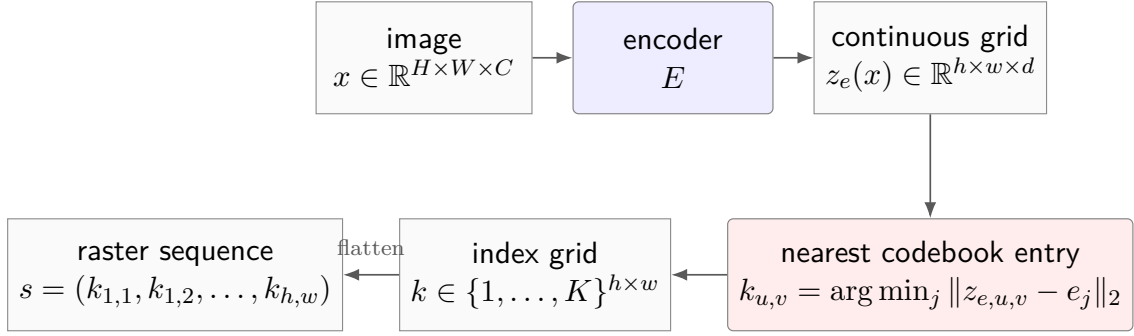
$$\hat{x} = D(z_q(x)). \quad (2.5)$$

Here, D is the decoder and $\hat{x}$ is the reconstruction. Figure 2.4 gives the corresponding geometric interpretation: codebook entries act as learned representative points in latent space.

The decoder receives $z_q(x)$ rather than the continuous encoder output. Image information must therefore pass through a finite vocabulary of K vectors. The same representation has two roles: the quantized vectors are the decoder input, while their integer indices form the discrete sequence modeled by the autoregressive prior. Figure 2.5 summarizes this conversion.

2.2.1. Learning the Discrete Representation

VQ-VAE is trained by gradient descent, but nearest-neighbor assignment is not differentiable. It therefore uses a straight-through estimator. During the forward pass the decoder receives quantized vectors; during backpropagation, the gradient with respect



Example dimensions for the evaluated checkpoints, not a universal tokenizer constraint: $h = w = 16$ and $K = 16,384$. The grid has 256 positions, and each token is one of 16,384 codebook indices.

Figure 2.5: From a continuous feature map to a discrete sequence. The encoder produces a latent grid, vector quantization replaces each latent vector with a codebook index, and the token grid is flattened for sequence modeling.

to the quantized representation is copied to the encoder output [4, 34]. The encoder, decoder, and codebook are learned together.

The objective contains a reconstruction loss and two additional terms for learning the codebook and encoder representation:

$$\mathcal{L}_{\text{VQ}} = \underbrace{-\log p(x | z_q(x))}_{\mathcal{L}_{\text{rec}}} + \underbrace{\|\text{sg}[z_e(x)] - z_q(x)\|_2^2}_{\mathcal{L}_{\text{cod}}} + \underbrace{\beta \|z_e(x) - \text{sg}[z_q(x)]\|_2^2}_{\mathcal{L}_{\text{comm}}}. \quad (2.6)$$

The reconstruction term follows the likelihood convention in the original VQ-VAE formulation [34]. The stop-gradient operator $\text{sg}[\cdot]$ is the identity in the forward pass and has zero derivative. The codebook term $\mathcal{L}_{\text{cod}}$ moves embeddings toward encoder outputs. The commitment term $\mathcal{L}_{\text{comm}}$ keeps encoder outputs close to their assigned embeddings.

The codebook contains K entries, but a dataset may activate only a subset. Codebook collapse occurs when many entries receive few or no assignments, which reduces the empirical vocabulary available to the autoregressive model [19, 21, 39]. Chapter 4 measures nominal and empirical vocabulary directly.

Nearest-neighbor assignment also partitions latent space into discrete regions. A small change in an encoder output can cross a region boundary and change its token abruptly, as illustrated in Figure 2.6.

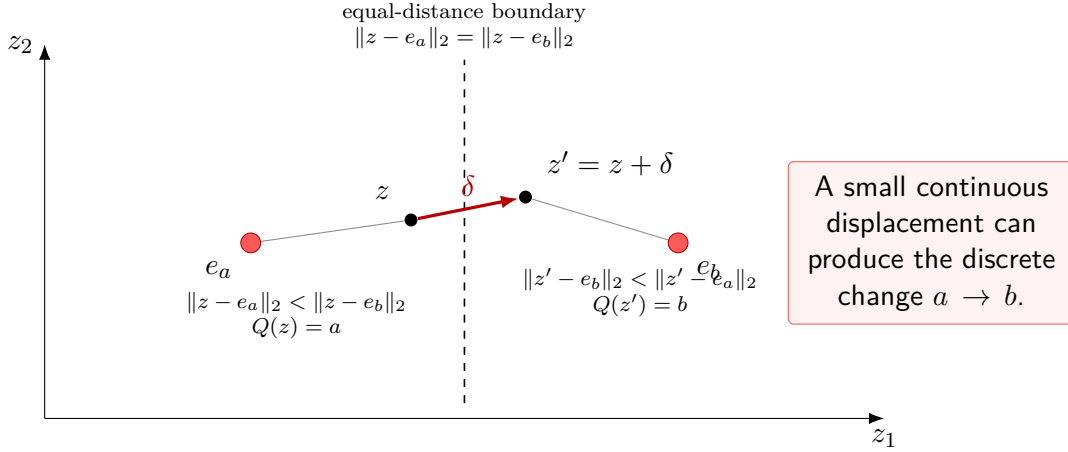


Figure 2.6: Nearest-neighbor regions create discontinuities in latent space. A small displacement of an encoder output can cross a quantization boundary and change its assigned token.

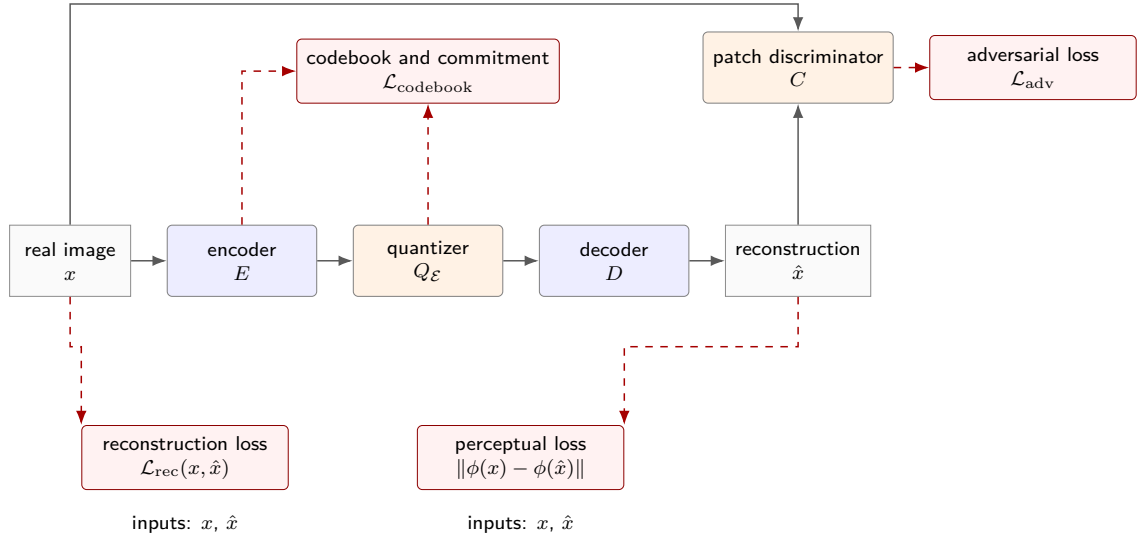
2.3. VQGAN and Taming Transformers

VQ-VAE showed how to learn a discrete token space that an autoregressive model can model, with a decoder that converts sampled token sequences into images [34]. Its reconstruction objective, however, does not necessarily preserve perceptually important high-frequency detail. When several sharp reconstructions are plausible, minimizing an average pixel error can favor a blurred result. Esser et al. address this limitation with a vector-quantized generative adversarial network, commonly called VQGAN, and use its discrete representation to make transformer-based high-resolution synthesis practical [10].

VQGAN retains the encoder–quantizer–decoder structure of VQ-VAE but changes the training objective. In the implementation used by the evaluated VQGAN checkpoint, the generator objective has four components:

$$\mathcal{L}_{\text{VQGAN}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{perc}} + \lambda \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{codebook}}. \quad (2.7)$$

This expanded implementation view separates the pixel reconstruction, perceptual, adversarial, and codebook terms; their configured and adaptive weights are omitted here for readability. The perceptual term compares activations of a pretrained network rather than requiring exact pixel agreement. A patch-based discriminator supplies the adversarial term. These objectives encourage the decoder to preserve perceptually salient structure and synthesize plausible fine detail. Figure 2.7 shows how the losses supervise the reconstructed image.



$$\mathcal{L}_{\text{VQGAN}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{perc}} + \lambda \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{codebook}}$$

Figure 2.7: VQGAN training combines reconstruction, perceptual, and adversarial supervision. Reconstruction and perceptual losses compare the input and reconstruction, while a patch discriminator encourages locally realistic outputs.

The first-stage model is convolutional, with non-local attention at its lowest-resolution level. This architecture combines local feature extraction with a limited mechanism for contextual interaction before quantization. Once trained, the tokenizer is frozen and images are represented by their codebook indices. A transformer is then trained autoregressively on the resulting sequences. The original architecture is reproduced in Figure 2.8.

A smaller token grid reduces the sequence length processed by the transformer, but forces each token to carry more image information and can reduce reconstruction fidelity. Esser et al. study this trade-off through different latent resolutions and two reported codebook sizes [10].

2.4. LlamaGen

After VQGAN, subsequent work improved autoregressive image generation. DALL-E, for example, models text and image tokens as a single autoregressive stream [26].

LlamaGen revisits this approach using a Llama-style transformer and reports competitive results against diffusion baselines through tokenizer and model design [31]. The system preserves the two-stage design: a vector-quantized tokenizer converts images to discrete grids, and an autoregressive model predicts their tokens.

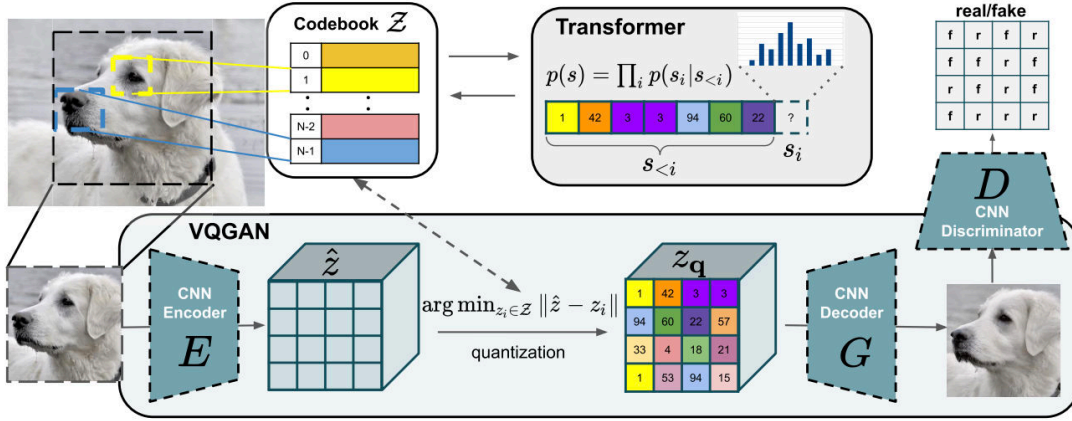


Figure 2.8: Taming Transformers architecture, comprising the VQGAN tokenizer, learned codebook, autoregressive transformer, and patch discriminator. Reproduced from Esser et al. [10].

A large nominal codebook is of limited value if the tokenizer uses only a small fraction of its entries. Following the factorized, L2-normalized codebook design of ViT-VQGAN [39], LlamaGen uses low-dimensional embeddings and normalized lookup [31]. Reducing the embedding dimension makes nearest-neighbor assignment less sparse, while normalization places encoder projections and codebook entries on a common unit sphere before assignment. LlamaGen also evaluates longer token sequences as a separate design choice. This thesis evaluates its VQ-16 tokenizer.

2.5. Representation Properties Beyond Reconstruction Quality

A tokenizer is usually evaluated first through reconstruction: given x , it produces $\hat{x} = D(Q(E(x)))$, and the two images are compared. Reconstruction is essential because any information discarded by the tokenizer is unavailable to the generative model. It is not, however, a complete description of the induced discrete representation.

The tokenizer also determines the effective vocabulary, token frequencies, and spatial response to perturbations. Two tokenizers with similar reconstruction quality may use different fractions of their codebooks. A small image perturbation may leave one token grid largely unchanged while altering many entries in another. A local token edit may remain local or propagate through the decoder. Chapters 4–8 measure these properties because the autoregressive model learns the distribution created by the tokenizer rather

than the original pixel distribution directly.

The two evaluated tokenizers share a VQGAN-style convolutional encoder and decoder but differ in their codebook projection, normalization, and training recipes [10, 31]. Their comparison describes the behavior of these two complete checkpoints. It cannot establish which design choice causes a difference or which behavior extends to all vector-quantized tokenizers. Earlier work on ViT-VQGAN and LlamaGen studied codebook use and reconstruction [31, 39], and EdiBERT showed qualitatively that changing a local image region can alter VQGAN tokens beyond that region [16]. TokenSet also reports Gaussian-noise token overlap for a VQGAN baseline under a different protocol and uses that comparison to motivate a set tokenizer [12]. Qiu et al. replace tokens with nearby codebook entries during tokenizer training to model sampling errors [25]; that is a precedent for studying near-code edits, not the same as measuring the spatial response of an unchanged decoder. The contribution here is a common evaluation of two specified frozen checkpoints: full-validation-set global and local perturbation responses, encoder distance profiles, decoder edit responses, and code-frequency redistribution across datasets in Chapters 5–8.

3 | Datasets, Models, and Metrics

This work focuses on two pretrained image tokenizers: VQGAN [10] and LlamaGen VQ-16 [31]. This chapter defines these two tokenizers, the evaluation datasets, and the metrics used to measure reconstruction quality, codebook use, and spatial behavior.

3.1. Models Under Study

These two tokenizers share the same external task, mapping 256×256 images to discrete token grids. They use the same VQGAN-style convolutional encoder-decoder backbone, while their quantizer projections, codebook geometry, and training recipes differ [10, 31].

VQGAN uses an encoder, vector quantizer, and decoder with a convolutional architecture and attention at the lowest-resolution level [10]. Section 2.3 describes the model in more detail. The pretrained version used in this study was trained with a reconstruction, perceptual, adversarial, and codebook objective. It has a downsampling factor $f = 16$, so a 256×256 image becomes a 16×16 token grid. Its codebook has $K = 16,384$ entries with dimension 256.

LlamaGen provides two pretrained versions, VQ-16 and VQ-8, with downsampling factors of 16 and 8, respectively [11, 31]. To maintain the same token-grid size as VQGAN, these experiments use VQ-16. It maps 256×256 images to a 16×16 grid with a nominal codebook of $K = 16,384$. Its codebook embeddings have dimension $d = 8$. Before nearest-neighbor lookup, the projected encoder vectors and codebook vectors are normalized to unit L2 norm.

Table 3.1 summarizes the configurations of the two evaluated tokenizers. Both implementations use channel multipliers (1, 1, 2, 2, 4), two residual blocks per encoder scale, three per decoder scale, GroupNorm with 32 groups, and self-attention at the coarsest spatial level and middle blocks. The quantizer projection and codebook geometry distinguish the implementations without requiring a different spatial backbone.

Property	VQGAN	LlamaGen
Source	Taming Transformers [10]	LlamaGen [31]
Main tokenizer	ImageNet first stage, $f = 16$	VQ-16
Input resolution	256×256	256×256
Token grid	16×16	16×16
Encoder/decoder	VQGAN-style convolutional backbone	VQGAN-style convolutional backbone
Nominal codebook size	$K = 16,384$	$K = 16,384$
Embedding dimension	256	8
Codebook normalization	none	L2-normalized lookup
Training loss components	pixel, perceptual, adversarial, codebook/commitment	pixel, perceptual, adversarial, codebook/commitment

Table 3.1: Tokenizer configurations compared in the thesis. Both models produce the same token-grid size and nominal vocabulary size and share the encoder–decoder backbone. LlamaGen uses an L2 pixel loss with LPIPS and PatchGAN terms [31]; the objectives differ in details and weights, rather than in the presence of perceptual or adversarial supervision.

3.2. Datasets

The ImageNet-1K validation split is used for reconstruction baselines, codebook usage, global perturbation robustness, and locality experiments. It belongs to the training domain of both selected checkpoints and appears in their published evaluations [10, 31]. ImageNet-V2 is also used in Chapter 7 for decoder locality. Chapter 8 studies distribution shifts using ImageNet-V2, ImageNet-Sketch, ObjectNet, BloodMNIST, OrganAMNIST, and RVL-CDIP.

3.2.1. ImageNet

ImageNet was introduced as a large-scale visual recognition dataset organized around the WordNet noun hierarchy [8]. It was collected to provide broad object-category coverage at a scale that earlier recognition datasets did not offer. The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) later standardized a 1,000-class classification benchmark, commonly called ImageNet-1K [29].

ImageNet is a standard reference dataset for the evaluated VQ tokenizers [10, 31]. We use the ImageNet-1K validation split because it matches their training domain. It contains 50,000 images, with 50 images for each of 1,000 object classes. The images are converted to RGB, resized, center-cropped to 256×256 , and normalized for each tokenizer. No test-time data augmentation is used.

3.2.2. Distribution-Shifted Datasets

These datasets evaluate the tokenizers beyond their training domain. We distinguish ImageNet-aligned shifts from datasets with different image content. ImageNet-V2 supports the decoder-locality analysis in Chapter 7; all datasets in this section support the reconstruction and codebook analyses in Chapter 8.

ImageNet-aligned shifts. ImageNet-V2 was collected years after the original ImageNet construction as a new test set for the same 1,000 classes [28]. It keeps the label space and natural-image setting close to ImageNet while changing the sampled images, so it represents a near-domain shift. The experiments use the 10,000-image matched-frequency variant in WordNet-ID directory format, not the threshold-0.7 or top-images variants.

ImageNet-Sketch instead keeps the ImageNet label space but replaces natural photographs with black-and-white sketches [36]. It represents a stronger visual change than ImageNet-

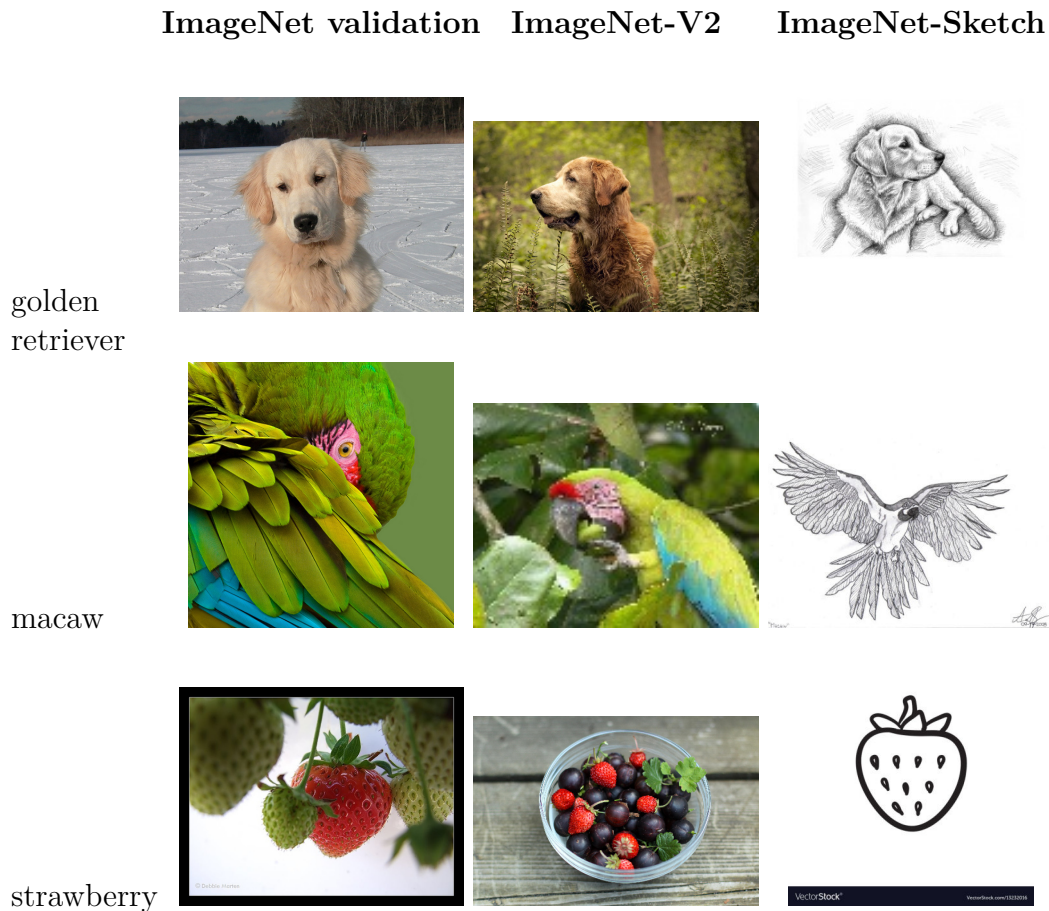


Figure 3.1: Examples from the ImageNet-aligned shift ladder. Each row keeps the same class while changing the collection process or visual style.

V2 because texture, color, and photographic detail are largely absent while object shape remains central. Figure 3.1 illustrates the three collections.

Non-ImageNet domain checks. Chapter 8 also uses datasets whose visual domains are not ImageNet-like. ObjectNet contains household objects photographed under controlled variations in viewpoint, rotation, and background, and was designed to expose object-recognition brittleness outside standard ImageNet biases [1]. BloodMNIST and OrganAMNIST are part of MedMNIST v2, a lightweight benchmark suite for biomedical image classification [38]: BloodMNIST contains RGB microscopy images of blood cells, while OrganAMNIST contains abdominal CT slices. RVL-CDIP contains scanned document pages labeled by document type [13]. These domains test tokenizer behavior on microscopy, medical images, and document layouts. Table 3.2 summarizes the datasets, and Figure 3.2 shows representative inputs.

Dataset	Content	Images used	Role in Chapter 8
ObjectNet	Controlled object photos with varied viewpoints and backgrounds	50,273	Natural-object domain with different pose and background statistics.
BloodMNIST	Microscopic RGB blood-cell images, 8 classes	17,092	Biomedical microscopy, dominated by local texture and cellular morphology.
OrganAMNIST	Abdominal CT slices, 11 organ classes	57,187	Grayscale medical imaging with intensity statistics far from natural images.
RVL-CDIP	Scanned document pages, 16 document classes	39,996	Layout- and text-like structure rather than natural-image content.

Table 3.2: Non-ImageNet collections evaluated. Counts describe the saved inputs; RVL-CDIP contains 400,000 images in the full source dataset.

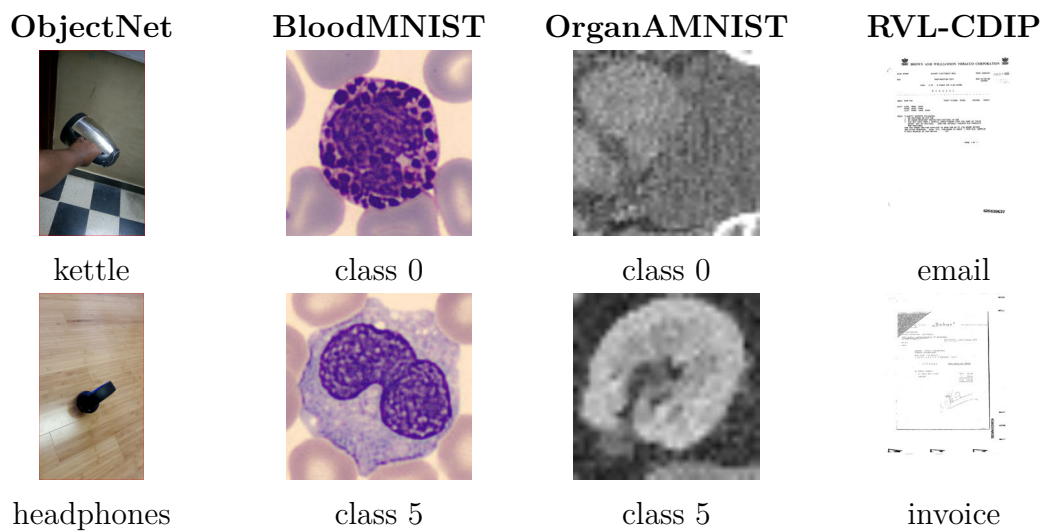


Figure 3.2: Illustrative non-ImageNet inputs selected to show the different content types.

The MedMNIST inputs are the 224-pixel exports, with train, validation, and test images pooled for this unsupervised tokenizer analysis. BloodMNIST contributes 11,959 training, 1,712 validation, and 3,421 test images. The local OrganAMNIST export contains 32,918 training, 6,491 validation, and 17,778 test images. The official release has 34,561 training images: 1,643 are absent from the saved PNG collection, with no recorded sampling rule. We therefore cannot assume the missing images are a random subset. Grayscale OrganAMNIST images are converted to three RGB channels before tokenization. For RVL-CDIP, the test export found 39,997 files and skipped one unreadable TIFF, leaving 39,996 images. ObjectNet uses the local 1.0 image collection. Its source images have a red identification border [1]; the exporters did not include a border-removal step. Results for ObjectNet and OrganAMNIST apply to these saved exports.

Metric family	Quantities used	Comparison unit	Interpretation
Paired-image reconstruction	PSNR, SSIM, LPIPS	One input x and its reconstruction $\hat{x}$	PSNR, SSIM $\uparrow$; LPIPS $\downarrow$
Image-set distributions	FID, sFID, precision, recall, IS	Reference and reconstruction sets	distribution agreement; IS is diagnostic
Codebook usage	Active codes, entropy, perplexity, top- m mass	All token assignments in an evaluation set	descriptive, not a quality score
Perturbation locality	Flip rate, inside/outside response, leakage ratio	Matched outputs and spatial regions	less outside response means greater locality

Figure 3.3: Metric taxonomy used in this thesis. Evaluation combines sample-level reconstruction metrics, distribution-level image metrics, codebook-usage measures, and locality-oriented measures.

3.3. Metrics

A tokenizer can be evaluated by reconstruction quality, codebook usage, and its response to perturbations. Token frequencies determine the distribution that the autoregressive model must learn. Perturbation response can be measured by how far changes spread through the token grid or decoded image.

We group the metrics by what they measure: paired-image reconstruction, image-set distributions, codebook usage, and perturbation locality. Paired metrics compare each reconstruction with its source image. Distribution metrics compare the aggregate distributions of two image sets; FID and related metrics perform this comparison in a particular learned feature space. Codebook metrics measure which codes are used and how concentrated their frequencies are. Perturbation metrics compare matched outputs before and after an intervention, either globally or within specified spatial regions. Figure 3.3 summarizes these four families.

3.3.1. Sample-Level Reconstruction Metrics

Full-reference metrics compare each reconstruction with its corresponding original image. Mean squared error (MSE) and peak signal-to-noise ratio (PSNR) are full-reference pixel metrics. For an image with N scalar pixel values,

$$\text{MSE}(x, \hat{x}) = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2, \quad (3.1)$$

$$\text{PSNR}(x, \hat{x}) = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}(x, \hat{x})} \right). \quad (3.2)$$

For normalized images in $[0, 1]$, $\text{MAX} = 1$. Lower MSE and higher PSNR indicate better pixel agreement. PSNR is widely used in image restoration but is known to be limited as

a perceptual quality measure [15].

SSIM compares local luminance, contrast, and structure rather than only pointwise pixel error [37]. In simplified form,

$$\text{SSIM}(x, y) = l(x, y)c(x, y)s(x, y), \quad (3.3)$$

where l , c , and s denote luminance, contrast, and structure comparisons computed over local windows. SSIM values are then averaged across the image. Higher SSIM indicates stronger structural similarity.

LPIPS compares normalized deep-network activations between two images [40]:

$$\text{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \left\| w_l \odot \left(\hat{\phi}_l(x)_{h,w} - \hat{\phi}_l(y)_{h,w} \right) \right\|_2^2, \quad (3.4)$$

where $\hat{\phi}_l$ denotes channel-normalized features at layer l , w_l contains learned channel weights, and the distance is averaged over the $H_l \times W_l$ spatial locations. Lower LPIPS indicates greater perceptual similarity. Unlike MSE and PSNR, LPIPS is calibrated against human perceptual judgments. The reconstruction evaluator used here uses its VGG-16 backbone.

3.3.2. Distribution-Level Image Metrics

FID compares two image sets by embedding them with an Inception network and fitting a Gaussian to each feature distribution [14]. For reference features with mean and covariance (μ_r, Σ_r) and reconstruction features with (μ_g, Σ_g) ,

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}). \quad (3.5)$$

Lower FID indicates closer feature distributions. In the literature on VQ tokenizers, VQGAN and LlamaGen report reconstruction FID or rFID to compare the distribution of reconstructions with original images [10, 31].

sFID uses spatial Inception features, following Nash et al. [22], and is implemented in the guided-diffusion evaluator [9, 23]. It retains the finite-sample and feature-extractor caveats of FID while measuring spatial features rather than only pooled features.

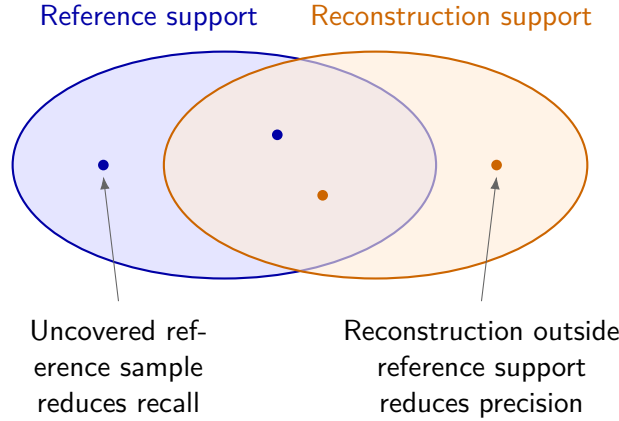


Figure 3.4: Schematic of the fidelity–coverage distinction. Precision concerns generated samples near the reference support, whereas recall concerns reference samples covered by the generated support. The diagram is conceptual, not the metric computation; see Kynkäänniemi et al. [18] for the formal method. *Source: author-created with AI assistance.*

Inception Score uses an Inception classifier [30]:

$$\text{IS} = \exp(\mathbb{E}_{x \sim p_g} [\text{KL}(p(y | x) \| p(y))]) . \quad (3.6)$$

Here, $p(y | x)$ is the classifier’s conditional class distribution for one reconstruction and $p(y)$ is the marginal class distribution across the set. The score increases when individual predictions are confident and predictions across the set cover different classes. It does not measure whether a reconstruction matches its input, so we report it as a classifier-based diagnostic without assigning it a preferred reconstruction direction [2, 6].

For reconstructed image sets, precision measures how much of the reconstruction distribution lies near the reference feature distribution, while recall measures how much of the reference distribution is covered [18]. Figure 3.4 illustrates this distinction.

3.3.3. Codebook Usage Metrics

Let n_k be the number of assignments to code k over an evaluation set, let $N_t = \sum_{k=1}^K n_k$ be the total number of token assignments, and let

$$p_k = \frac{n_k}{N_t} . \quad (3.7)$$

p_k is the fraction of all observed token assignments that use code k . The raw active-code count is

$$A_{\text{raw}} = \sum_{k=1}^K \mathbf{1}[n_k > 0], \quad (3.8)$$

and the active fraction is

$$a_{\text{raw}} = \frac{A_{\text{raw}}}{K}. \quad (3.9)$$

The active-code count is the number of distinct codes used at least once; dividing it by K gives the active fraction. Because a single assignment makes a code active under this definition, Chapter 4 also reports two support measures. The effective active set counts codes with $p_k \geq 10^{-6}$, which is at least 13 of the 12.8 million assignments in that evaluation. The cumulative support C_{99} is the smallest frequency-ranked set of codes covering 99% of assignments. These measures are distinct from perplexity, which summarizes the full frequency distribution.

Entropy and perplexity summarize how spread out the token distribution is:

$$H(p) = - \sum_{k=1}^K p_k \log p_k, \quad (3.10)$$

$$P(p) = \exp(H(p)). \quad (3.11)$$

The intuitive meaning of perplexity is effective vocabulary size. If the tokenizer used exactly M codes uniformly, then $P(p) = M$. A tokenizer may therefore have $K = 16,384$ nominal entries but a much smaller effective vocabulary if the distribution is concentrated.

Top- m mass measures concentration directly:

$$M_m = \sum_{k \in \text{Top}_m(p)} p_k, \quad (3.12)$$

where $\text{Top}_m(p)$ is the set of the m most frequent codes. High top- m mass means that a small set of codes accounts for a large fraction of assignments.

Finally, positional entropy computes the same entropy quantity at each grid position (u, v) across the dataset. It measures code diversity at that position. Equal entropy at two positions does not imply that they use the same code identities; comparing those identities requires an overlap or distribution-divergence measure.

3.3.4. Locality and Robustness Metrics

Perturbation metrics compare tokenizer outputs before and after a controlled change. A global response can be measured as the change in reconstruction or codebook statistics. Spatial locality uses a mask to divide the response into inside and outside regions. For token position i , define a flip indicator

$$F_i = \mathbf{1}[t_i \neq t'_i], \quad (3.13)$$

where t_i and t'_i are the clean and perturbed token assignments. For a set of token positions R ,

$$r(R) = \mathbb{E}_{i \in R}[F_i]. \quad (3.14)$$

For a local image perturbation, the leakage ratio compares outside-region and inside-region flip rates:

$$\lambda_{\text{enc}} = \frac{r(O_P)}{r(I_P) + \varepsilon}. \quad (3.15)$$

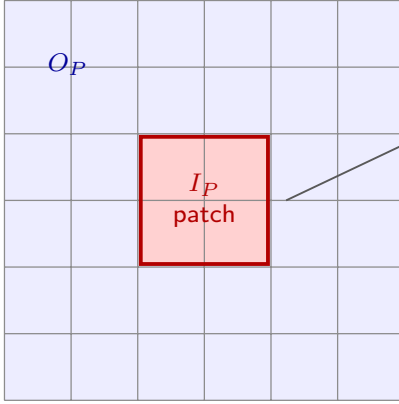
Here I_P and O_P are the token positions inside and outside patch P , and $\varepsilon = 10^{-8}$. The rates are denoted $R_{\text{in}} = r(I_P)$ and $R_{\text{out}} = r(O_P)$ in Chapter 6. The ratio compares response densities, not total numbers of changed tokens; distance profiles are needed to establish how far the response extends.

The same definition applies to decoder locality after replacing the token flip indicator with an image-change measure. Later chapters specify the perturbation, mask, and response measure used by each experiment. Figure 3.5 shows the region partition.

3.3.5. Metric Comparison Summary

Table 3.3 summarizes the metrics used in the thesis. Paired metrics compare images, distribution metrics compare image sets, codebook metrics describe token frequencies, and locality metrics describe the response to controlled perturbations.

spatial response grid

**Flip rate in a region**

$$F_i = \mathbf{1}[t_i \neq t'_i], \quad r(R) = \frac{1}{|R|} \sum_{i \in R} F_i$$

Encoder leakage ratio

$$\lambda_{\text{enc}} = \frac{r(O_P)}{r(I_P) + \varepsilon}$$

Interpretation

The ratio compares outside and inside response densities. Decoder locality uses the analogous inside/outside partition in image space.

Figure 3.5: Locality and leakage as measured in this thesis. Encoder locality compares clean and perturbed token assignments inside and outside the image patch mapped to token space. Decoder locality uses the analogous partition of the decoded image response.

Metric	What it measures	Level	Direction	Main caveat
MSE / PSNR	Aligned pixel error	Sample image pair	MSE ↓, PSNR ↑	Poor proxy for perceptual similarity.
SSIM	Local luminance, contrast, and structure	Sample image pair	↑	Sensitive to alignment and window choices.
LPIPS	Deep-feature perceptual distance	Sample image pair	↓	Depends on backbone and calibration.
FID / sFID	Feature-distribution match between image sets	Image set	↓	Feature extractor, finite-sample, and Gaussian-assumption effects.
IS	Classifier confidence and predicted-class diversity	Image set	descriptive	Does not directly compare outputs to real images.
Precision / Recall	Fidelity to and coverage of a reference feature manifold	Image set	both ↑	Depends on feature space and neighborhood construction.
Active count / fraction	How much of the nominal codebook appears	Token distribution	descriptive	Raw activity is sensitive to rare one-off codes.
Entropy / Perplexity	Effective spread of code usage	Token distribution	descriptive	Can hide spatial specialization.
Top- m mass	Concentration in frequent codes	Token distribution	descriptive	Choice of m must be reported.
Flip rate / leakage ratio	Token or image changes under controlled perturbation	Perturbation response	descriptive	Depends on perturbation mask and region definition.

Table 3.3: Compact comparison of metric families used in the thesis. Arrows indicate a preferred direction for reference-based fidelity metrics under a fixed protocol. Other metrics describe behavior rather than a universal ranking.

4 | Reconstruction Baseline and Codebook Usage

Before any perturbation is introduced, this chapter establishes the clean ImageNet baseline for the two studied tokenizers. It measures how well each tokenizer reconstructs unmodified validation images and which empirical token distribution it produces. Both tokenizers have the same nominal vocabulary size, $K = 16,384$, and produce a 16×16 token grid from a 256×256 image, making their outputs directly comparable.

4.1. Experimental Setup

For each image x , the tokenizer produces a reconstruction:

$$\hat{x} = D(Q(E(x))), \quad (4.1)$$

where E is the encoder, Q is the vector-quantization step, and D is the decoder. Reconstruction quality is measured using the metrics defined in Chapter 3: PSNR, SSIM, LPIPS, FID, sFID, precision, recall, and Inception Score. Codebook usage is computed from the discrete token grids over all 50,000 validation images. Since each image produces a 16×16 grid, each tokenizer contributes:

$$50,000 \times 16 \times 16 = 12,800,000 \quad (4.2)$$

token assignments for the ImageNet baseline.

The codebook analysis reports both raw and effective vocabulary use. The raw active count is the number of codebook entries assigned at least once. The effective active count uses a minimum-frequency threshold of 10^{-6} of all token assignments, corresponding to at least 13 assignments out of 12.8 million tokens in this experiment. Entropy and perplexity summarize the effective size of the empirical distribution, and top- m mass measures how

Tokenizer	PSNR	SSIM	LPIPS	FID	sFID	Precision	Recall	IS
LlamaGen	20.79	0.558	0.228	2.19	5.01	0.978	0.995	52.70
VQGAN	19.65	0.489	0.286	5.06	6.49	0.920	0.973	46.76

Table 4.1: ImageNet reconstruction metrics for the two tokenizers under their recorded preprocessing. PSNR is in dB. Higher PSNR, SSIM, precision, and recall are preferred; lower LPIPS, FID, and sFID are preferred.

much of the token stream is captured by the most frequent codes.

The experiment uses the same ImageNet validation images and deterministic inference for both models. Each exporter applies its own resize filter before the 256×256 center crop; Appendix B documents this preprocessing difference.

4.2. Reconstruction Baseline

Table 4.1 reports the ImageNet reconstruction metrics for LlamaGen and VQGAN. LlamaGen has better scores on the paired-image metrics (PSNR, SSIM, LPIPS) and the reference-conditioned distribution metrics (FID, sFID, precision, and recall). Inception Score is reported alongside them as the diagnostic defined in Chapter 3.

The PSNR gap is modest in absolute terms, but it is consistent with the other metrics. LlamaGen reaches a mean PSNR of 20.79 compared with 19.65 for VQGAN, and a mean SSIM of 0.558 compared with 0.489. LPIPS also favors LlamaGen: 0.228 compared with 0.286 for VQGAN. Because LPIPS is computed in a learned feature space, the reconstruction difference extends beyond pixel-level measurements. The distributional metrics agree with the same direction: LlamaGen obtains FID 2.19 and sFID 5.01, while VQGAN obtains FID 5.06 and sFID 6.49.

The corresponding per-image standard deviations are 3.18 dB, 0.167, and 0.058 for LlamaGen PSNR, SSIM, and LPIPS, and 3.41 dB, 0.176, and 0.066 for VQGAN. These standard deviations describe variation across images, not uncertainty in the dataset means. A separate clean-condition export from the robustness run supplies PSNR for both models on the same 50,000 ImageNet validation IDs. Its model means differ from the reconstruction export above by less than 0.001 dB. Pairing those rows by image, LlamaGen’s mean PSNR advantage is 1.144 dB, with a 95% image-level bootstrap percentile interval of [1.140, 1.148] dB from 2,000 resamples. LlamaGen has higher PSNR on 97.91% of images. The interval describes sampling variation across validation images, not uncertainty from preprocessing or checkpoint choice. We do not have corresponding paired per-image SSIM or LPIPS rows from this export. FID and related distribution metrics do not decompose

Tokenizer	Codebook size		Active	Effective active	Perplexity	Top-10	Top-100	Top-500
LlamaGen	16,384	16,384 (100.0%)		16,384 (100.0%)	16,179.19	0.003	0.012	0.046
VQGAN	16,384	972 (5.9%)		971 (5.9%)	910.46	0.022	0.165	0.665

Table 4.2: ImageNet codebook usage summary. Active counts report how many nominal codes appear on ImageNet; effective active counts use the threshold of at least 13 assignments. Top- m values report the fraction of all token assignments covered by the most frequent m codes.

into per-image observations and are reported as dataset-level estimates.

The published LlamaGen comparison reports PSNR 20.79, SSIM 0.675, and reconstruction FID 2.19 for VQ-16, versus PSNR 20.00, SSIM 0.629, and FID 4.99 for VQGAN-f16-16384 [31]. Our LlamaGen PSNR and FID agree to the displayed precision. The SSIM values are not directly comparable. The public LlamaGen evaluation script sets the SSIM data range to 2.0 for images in $[0, 1]$, whereas our exporters set it to 1.0 and use scikit-image’s default window with the last axis as the color channel. Our VQGAN PSNR and FID also differ from the published values; the resize filters differ between exporters, so these numbers do not establish the cause of that offset. LlamaGen’s published 97% usage is measured in a 65,536-token queue, whereas our 100% denotes codes observed at least once across 12.8 million assignments.

Figure 4.1 shows how much detail and structure each tokenizer preserves through the discrete bottleneck on three paired examples.

4.3. Codebook Usage

The metrics above show that LlamaGen reconstructs the ImageNet validation images more accurately than VQGAN under the recorded exporter settings. The following analysis examines whether the two tokenizers also differ in how they use their discrete vocabularies. Table 4.2 reports the main usage statistics.

LlamaGen uses all 16,384 codebook entries on ImageNet with a perplexity of 16,179. This means that the empirical token distribution is close to uniform over the full nominal vocabulary. VQGAN uses only 972 of 16,384 codes, or about 5.9% of the nominal codebook. Liu et al. [20] previously reported approximately 1,000 used codes for this released VQGAN checkpoint on ImageNet training data; our validation-set count confirms that underuse under a different sample and protocol. Its perplexity of 910 is close to the raw active count, so the active subset is small but not dominated by only a handful of codes.

LlamaGen’s top-500 codes account for only 4.6% of all token assignments; VQGAN’s

Tokenizer	PSNR	SSIM	LPIPS	FID	Active	Perplexity	Top-100	Codes for 99%
LlamaGen	20.79	0.558	0.228	2.19	16,384 (100.0%)	16,179.19	0.012	16,144
VQGAN	19.65	0.489	0.286	5.06	972 (5.9%)	910.46	0.165	946

Table 4.3: Joint ImageNet summary of reconstruction quality and codebook usage. The final column reports the number of most frequent codes required to cover 99% of token assignments.

top-500 account for 66.5%. Thus, although the two tokenizers expose the same nominal vocabulary size, the sequences they would provide to a downstream autoregressive model have very different empirical vocabularies. The downstream model is trained on this empirical token stream, not on the unused portion of the nominal codebook.

Entropy gives an idealized coding interpretation of the same result. Since $H = \log(P(p))$, the observed perplexities correspond to about 9.69 nats (13.98 bits) per token for LlamaGen and 6.81 nats (9.83 bits) per token for VQGAN. For a 256-token grid, these are approximately 3,579 and 2,517 idealized bits per image under an optimal code matched to each empirical marginal distribution. LlamaGen’s empirical marginal entropy is therefore about 42% higher. These marginal entropies describe ideal code lengths for the observed token frequencies. They do not measure semantic content or the conditional code length of a trained autoregressive model.

Figure 4.2 shows the full rank-frequency curve. LlamaGen assigns probability mass across the full codebook. VQGAN has a shorter curve because most nominal codes are inactive on ImageNet, and its active codes carry substantially higher average frequency.

The top- m columns in Table 4.2 report the cumulative mass of the 10, 100, and 500 most frequent codes. High top- m mass means that a large fraction of the token sequence comes from a small set of IDs.

4.4. Baseline Summary

The baseline has two parts. First, LlamaGen gives better clean ImageNet reconstructions under the reported sample-level and distribution-level metrics. Second, the two tokenizers use their nominally equal codebooks very differently. Table 4.3 keeps a compact version of these two facts together.

These results set the ImageNet baseline for later chapters. As expected from the newer tokenizer’s stated design goals, LlamaGen reconstructs the validation set more accurately and uses nearly its full codebook. VQGAN reconstructs less accurately and leaves roughly 94% of its nominal codes inactive. Such codebook underuse is a documented failure mode

in discrete representation learning [19, 21, 39], and LlamaGen explicitly targets it through its tokenizer design [31].



Figure 4.1: Illustrative ImageNet reconstruction examples, selected from the saved archive to show different content. They are not a random or median-quality sample. Each row shows the original image and the two reconstructions.

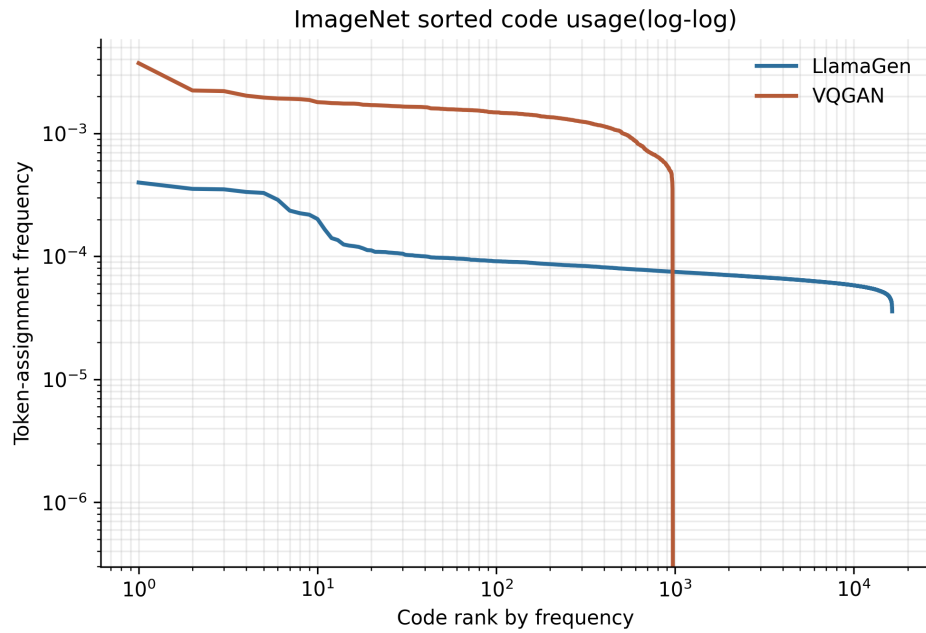


Figure 4.2: Sorted ImageNet code-usage distribution. Codes are ranked by empirical frequency. The VQGAN curve ends near rank 972 because the remaining nominal codes are inactive on ImageNet.

5 | Robustness to Global Perturbations

This chapter investigates how global Gaussian noise affects reconstructions and token assignments. We assess robustness through reconstruction quality and changes in token assignments, then aggregate assignments across the dataset to measure changes in code frequencies. The autoregressive model receives token sequences, so a reconstruction may remain recognizable even when many assignments change. Nearest-neighbor quantization can switch an assignment when an encoder output crosses a codebook boundary, although this experiment does not measure distances to those boundaries. Gaussian-noise token overlap has also been reported for a VQGAN baseline in TokenSet [12]. The present experiment uses the two specified checkpoints, reports reconstruction and frequency response as well as identity flips, and does not assume the papers’ noise scales or checkpoints coincide.

5.1. Experimental Setup

5.1.1. Perturbation Protocol

Let $x \in [-1, 1]^{H \times W \times 3}$ denote an input image in the tokenizer’s internal range. For a noise level σ , the perturbed image is defined by:

$$x_\sigma = \text{clip}(x + \sigma\epsilon, -1, 1), \quad \epsilon \sim \mathcal{N}(0, I). \quad (5.1)$$

Clipping keeps the perturbed image in the allowed range and can place values at its limits. The applied perturbation is therefore clipped Gaussian noise, not an unbounded Gaussian draw; the fraction of clipped values was not archived. The experiment uses $\sigma \in \{0.1, 0.2, 0.5\}$ together with the clean baseline $\sigma = 0$. In the reported $[0, 1]$ image range, these noise standard deviations are 0.05, 0.10, and 0.25 before clipping. The input corruption measured over all 50,000 images is reported in Table 5.1; only these three

nonzero levels were tested.

For a tokenizer with encoder E , quantizer Q , and decoder D , the perturbed reconstruction and tokenization are given by:

$$\hat{x}_\sigma = D(Q(E(x_\sigma))), \quad z_\sigma = \text{indices}(Q(E(x_\sigma))). \quad (5.2)$$

Applying the same operations to x gives the clean reconstruction $\hat{x}_0 = D(Q(E(x)))$ and clean token grid $z_0 = \text{indices}(Q(E(x)))$. Here Q returns selected embeddings, and indices records their codebook IDs. Each z_σ is a 16×16 grid.

The experiment was performed on the full ImageNet validation set for VQGAN and LlamaGen. Each image is evaluated clean and at $\sigma = 0.1, 0.2$, and 0.5 .

Both tokenizers use the same images and nominal noise levels, but their runs use different Gaussian noise realizations. Model comparisons are therefore matched by image and noise level, not by individual noise samples.

5.1.2. Measured Effects

Reconstruction fidelity. PSNR measures the input corruption, $\text{PSNR}(x, x_\sigma)$; fidelity to the noisy input, $\text{PSNR}(x_\sigma, \hat{x}_\sigma)$; and fidelity to the clean input, $\text{PSNR}(x, \hat{x}_\sigma)$. The last quantity is the main reconstruction measure in this chapter. None of these is $\text{PSNR}(\hat{x}_0, \hat{x}_\sigma)$. A separate fixed-seed sample of 2,000 images compares saved clean and noisy reconstruction PNGs directly. Those PNGs are eight-bit outputs, so the direct-change values include output quantization; the 50,000-image fidelity values above retain their original evaluation pipeline.

Token flip rate. The token flip rate is the fraction of assignments that differ from the clean tokenization:

$$\text{flip}_\sigma(x) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[z_{\sigma,i} \neq z_{0,i}], \quad (5.3)$$

where $N = 256$ is the number of positions in the 16×16 token grid. A value near zero indicates that most positions keep the same token under noise; a value near one indicates that nearly all token identities changed.

Flip rates are directly comparable across noise levels within a tokenizer. Their absolute values are only descriptive across tokenizers: LlamaGen uses almost the full nominal vocabulary, whereas VQGAN quantizes into a much smaller empirical subset, and the two codebooks induce different partitions of latent space. A higher cross-model flip rate

therefore does not by itself imply a worse representation.

Codebook usage. After measuring changes at individual positions, we aggregate assignments across images to examine code-frequency changes. For one empirical distribution p_k , the entropy and corresponding perplexity are given by:

$$H(p) = - \sum_k p_k \log p_k, \quad P(p) = \exp(H(p)). \quad (5.4)$$

Active count measures codebook coverage. Top- m mass measures how strongly assignments concentrate into a small number of codes.

5.2. Experimental Results

5.2.1. Qualitative Behavior

Figures 5.1 and 5.2 show selected ImageNet examples for the two tokenizers. Each example contains the clean image, inputs at $\sigma = 0.1, 0.2$, and 0.5 , the clean reconstruction, and the three perturbed reconstructions. These panels are qualitative examples; the quantitative conclusions use the full validation set.

5.2.2. Reconstruction Degradation

Table 5.1 reports $\text{PSNR}(x, \hat{x}_\sigma)$, the end-to-end fidelity of the noisy-input reconstruction to the original clean image. Figure 5.3 visualizes the same trend. Both tokenizers degrade monotonically as σ increases, and LlamaGen remains above VQGAN at every level. At the clean baseline, LlamaGen reaches 20.79 dB and VQGAN reaches 19.65 dB. At the highest perturbation level, LlamaGen remains at 16.22 dB, while VQGAN falls to 14.90 dB.

The clean reconstruction difference persists under the tested perturbations. The gaps are 1.14, 1.09, 1.00, and 1.32 dB from $\sigma = 0$ to 0.5 , so they are broadly similar rather than monotonically widening.

Figure 5.4 reports $\text{PSNR}(x_\sigma, \hat{x}_\sigma)$. At $\sigma = 0.1$, the values are 19.26 dB for LlamaGen and 18.36 dB for VQGAN, a 0.90 dB gap. At $\sigma = 0.5$, they are 11.87 and 11.31 dB, a 0.56 dB gap. LlamaGen has a modest advantage under this measure.

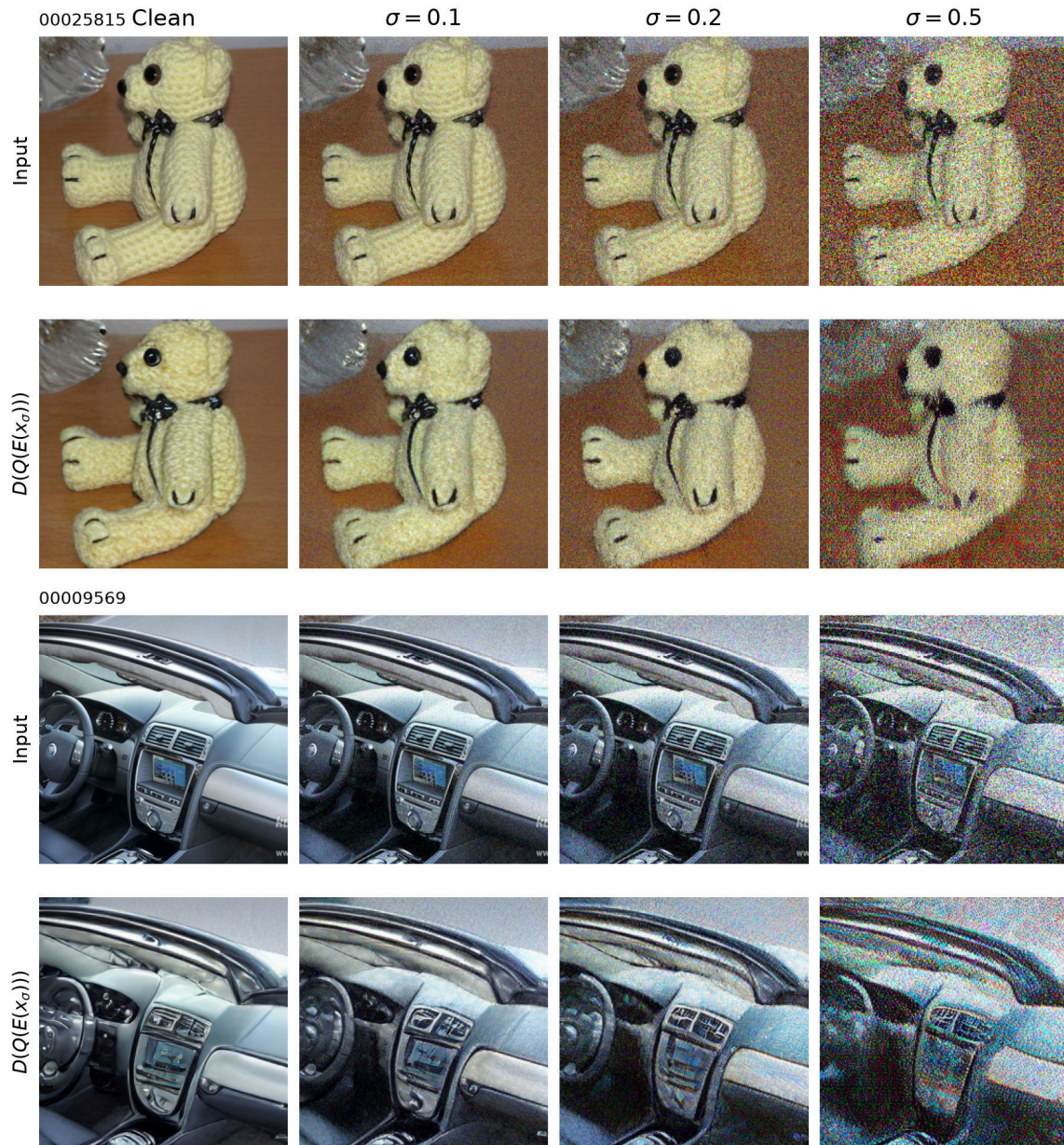


Figure 5.1: Selected VQGAN reconstructions under global Gaussian noise at $\sigma = 0.1, 0.2$, and 0.5 . The examples are illustrative; aggregate results use the full validation set.

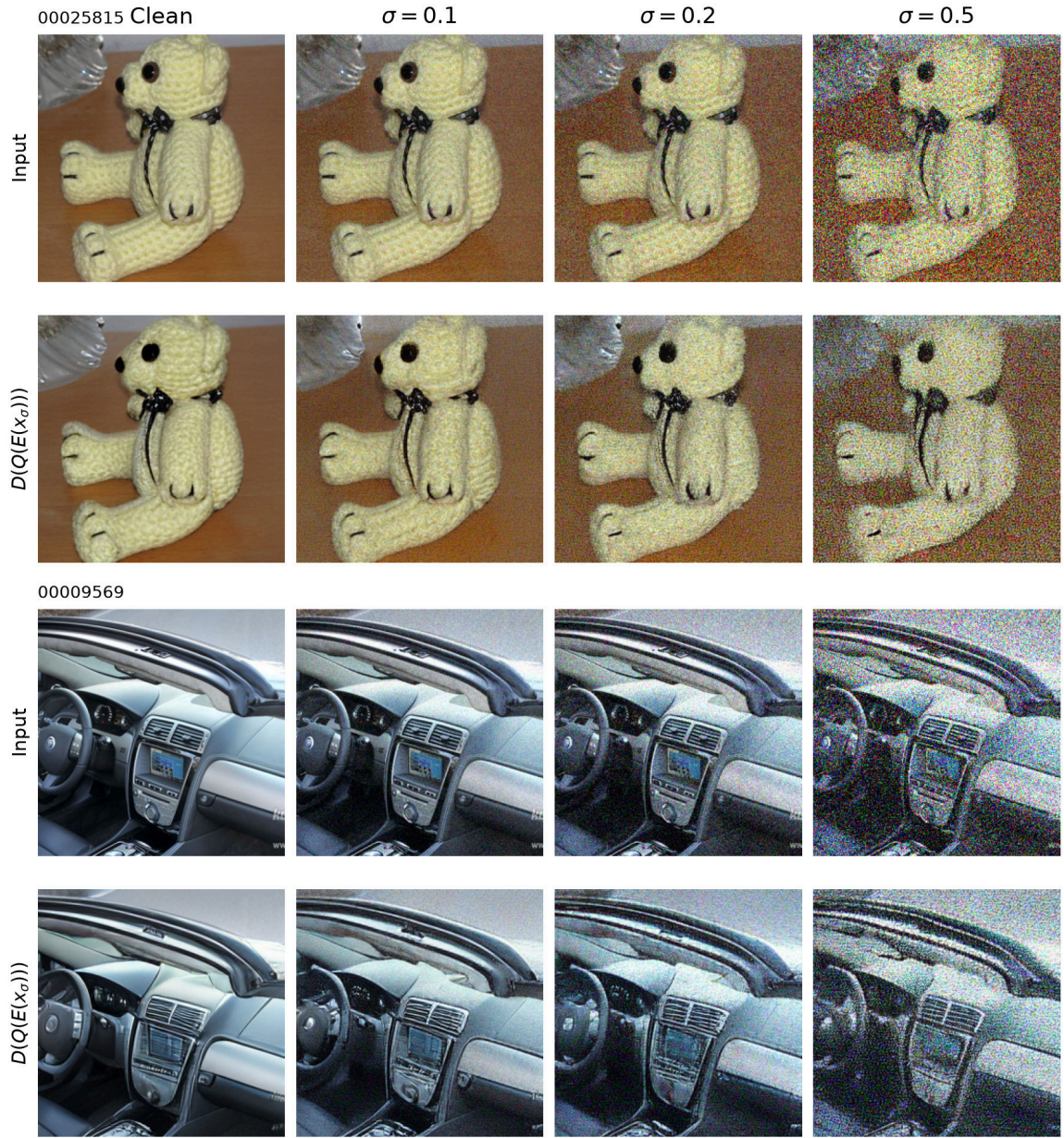


Figure 5.2: LlamaGen reconstructions for the same ImageNet examples and noise levels as Figure 5.1.

Tokenizer	Noise level	σ	PSNR(x, x_σ)	PSNR($x, \hat{x}_\sigma$)
LlamaGen	clean	0.0	—	20.79 ± 3.18
LlamaGen	low	0.1	26.29	20.32 ± 2.71
LlamaGen	mid	0.2	20.49	19.11 ± 1.98
LlamaGen	high	0.5	13.35	16.22 ± 0.94
VQGAN	clean	0.0	—	19.65 ± 3.41
VQGAN	low	0.1	26.30	19.23 ± 2.92
VQGAN	mid	0.2	20.50	18.11 ± 2.27
VQGAN	high	0.5	13.35	14.90 ± 1.03

Table 5.1: Input corruption and end-to-end reconstruction fidelity under global Gaussian perturbations. Values are mean PSNR in dB over 50,000 ImageNet validation images. The reconstruction column also shows per-image standard deviation; input PSNR is reported as a mean only. Input PSNR is undefined for the clean row because the images are identical.

Tokenizer	Noise level	σ	Token flip rate
LlamaGen	low	0.1	0.795 ± 0.099
LlamaGen	mid	0.2	0.910 ± 0.054
LlamaGen	high	0.5	0.979 ± 0.017
VQGAN	low	0.1	0.751 ± 0.103
VQGAN	mid	0.2	0.890 ± 0.057
VQGAN	high	0.5	0.975 ± 0.017

Table 5.2: Fraction of token positions changed relative to the clean tokenization. Values are mean $\pm$ standard deviation over ImageNet validation images.

5.2.3. Token Instability

The degradation in image space is gradual, while quantization can cause abrupt assignment changes. Table 5.2 and Figure 5.5 report the dataset mean of $\text{flip}_\sigma(x)$. At $\sigma = 0.1$, VQGAN changes 75.1% of token positions and LlamaGen changes 79.5%, even though the selected reconstructions shown in Figures 5.1 and 5.2 remain recognizable. At $\sigma = 0.2$, the mean flip rates rise to 89.0% and 91.0%; at $\sigma = 0.5$, both exceed 97%.

LlamaGen has higher clean-input reconstruction PSNR at every tested level. Its mean identity-flip rate is also slightly higher, but the codebooks induce different partitions and a cross-model flip-rate ranking is not a ranking of representation quality. These measurements describe fidelity to the original image and exact token identity; they do not determine how far a flipped code moves in embedding space. The saved-image comparison below measures decoded change separately.

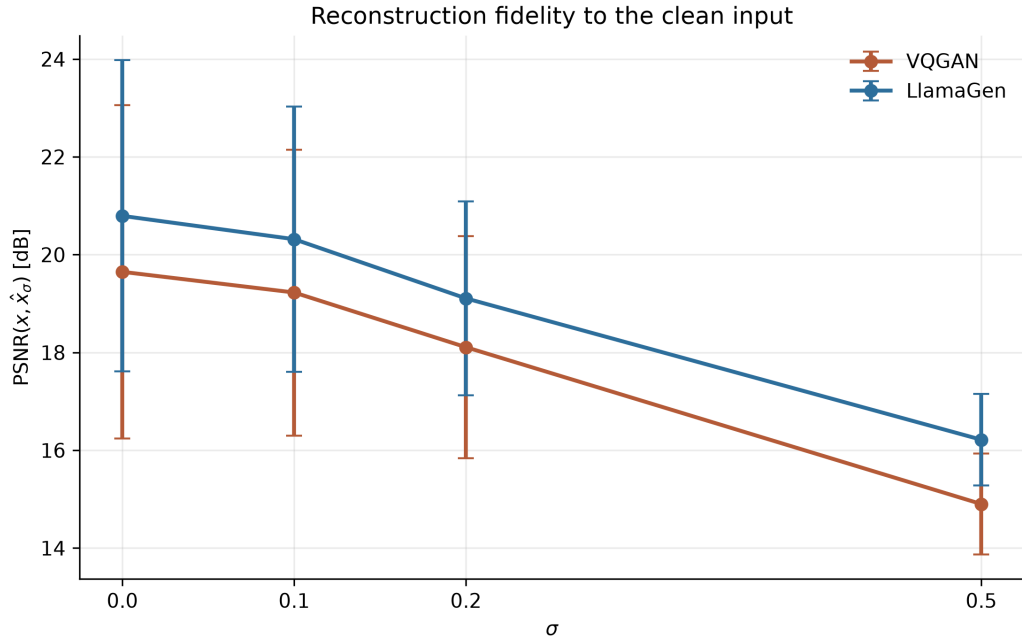


Figure 5.3: Reconstruction fidelity to the clean input. Points show mean $\text{PSNR}(x, \hat{x}_\sigma)$ and error bars show one standard deviation across images. The horizontal axis is σ and PSNR is in dB.

Direct decoded-image change. A separate check compares each noisy reconstruction with the clean reconstruction of the same input, rather than with the original image. The same 2,000 ImageNet validation IDs were selected for both models without replacement (seed 20260925), then read from the archived PNGs and token-index records; the two models’ noise realizations were not matched. Mean per-image decoded RMSE in $[0, 1]$ pixel units is 0.092/0.112/0.158 for LlamaGen and 0.099/0.122/0.182 for VQGAN at $\sigma = 0.1/0.2/0.5$, respectively. At $\sigma = 0.1$, LlamaGen has a higher identity-flip fraction than VQGAN in this sample (0.795 versus 0.751), but a lower decoded RMSE. The ordering holds at the two larger noise levels. Thus the identity-flip ordering does not carry over to decoded-image change in this sample; neither measure identifies the cause of the difference.

5.2.4. Global Codebook Distribution

After measuring changes at individual positions, we aggregate assignments across images to examine code-frequency changes. Table 5.3 summarizes active-code counts, perplexity, and top- m concentration; Figure 5.6 visualizes the main trend. Figure 5.7 compares clean and $\sigma = 0.5$ assignment frequencies for the same codes.

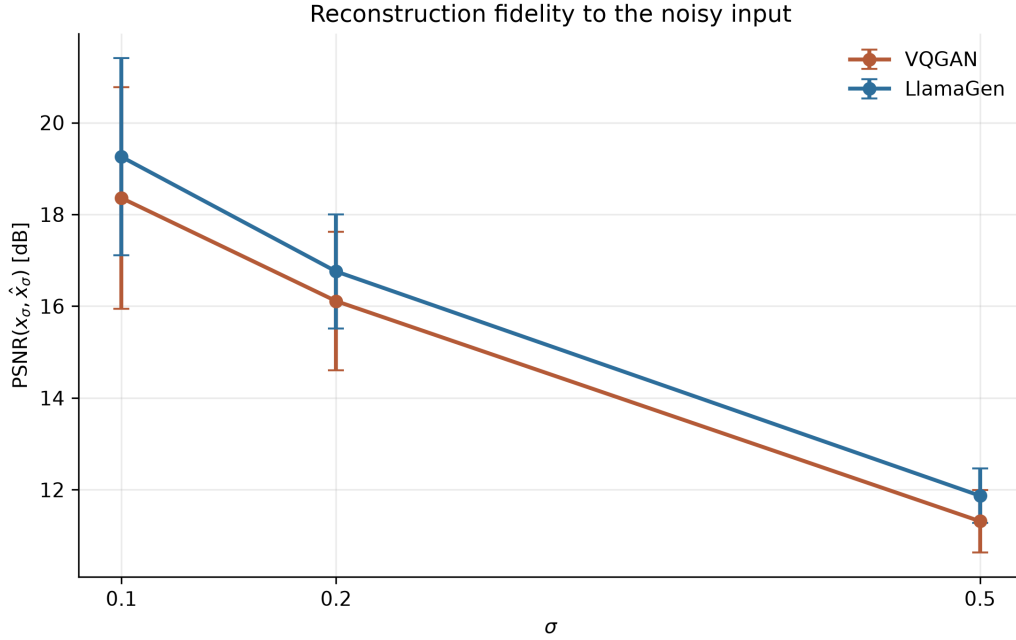


Figure 5.4: PSNR between noisy input x_σ and reconstruction $\hat{x}_\sigma$. This separates fidelity to the corrupted input from fidelity to the original clean image.

The two tokenizers have different baselines. LlamaGen uses the full codebook on clean ImageNet, with 16,384 active codes and a perplexity of 16,179. VQGAN uses 972 active codes, with a perplexity of 910. Under noise, LlamaGen’s active count remains high and reaches 16,203 at $\sigma = 0.5$; VQGAN’s stays near 972.

Perplexity is more sensitive than active-code count. LlamaGen’s perplexity decreases from 16,179 to 6,434 as σ increases from 0 to 0.5. VQGAN’s perplexity decreases from 910 to 560. Thus, both distributions become more concentrated even when the number of active codes changes little. The top-100 mass confirms this concentration: LlamaGen increases from 0.012 to 0.115, while VQGAN increases from 0.165 to 0.388. Top-1000 mass is omitted because VQGAN uses at most 973 codes at these levels, making that statistic identically one for VQGAN.

5.3. Discussion

Reconstruction and assignment response. The tokenizers show different clean-input reconstruction scores and high identity-flip rates under each tested perturbation. Their absolute flip rates cannot be used to rank stability across the two codebook geometries.

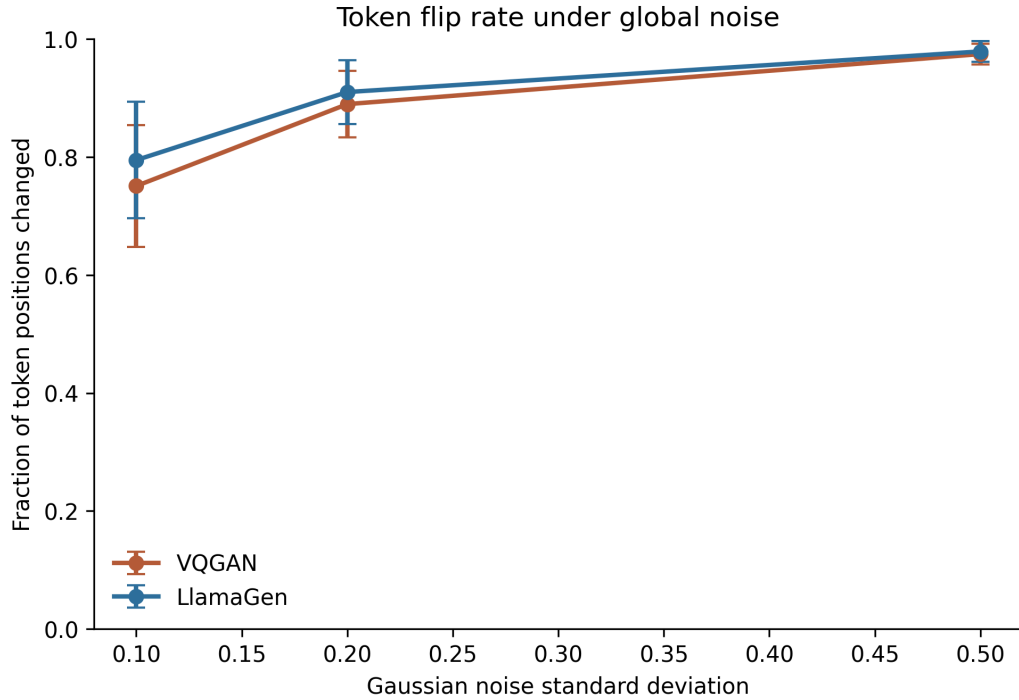


Figure 5.5: Token identity changes under global Gaussian perturbations.

The direct decoded-change sample gives the opposite checkpoint ordering from the identity-flip fractions. A changed code ID therefore has no fixed image-space magnitude across these checkpoints.

Code-frequency concentration. Perplexity decreases and top-100 mass increases for both tokenizers while active-code counts change comparatively little. The frequency distributions therefore become more concentrated under noise. Figure 5.7 shows this redistribution with code order fixed to the clean condition; the change cannot be explained by simply sorting the noisy frequencies again. The complete condition summaries are retained in Appendix C.

The results above provide the reference point for the locality analyses. Global noise perturbs all image regions. Chapter 6 studies a local image change, and Chapter 7 studies bounded token edits.

Tokenizer	Level	Active codes	Perplexity	Top-100 mass
LlamaGen	clean	16,384	16,179	0.012
LlamaGen	low	16,381	12,829	0.040
LlamaGen	mid	16,369	10,388	0.063
LlamaGen	high	16,203	6,434	0.115
VQGAN	clean	972	910	0.165
VQGAN	low	973	859	0.202
VQGAN	mid	971	780	0.249
VQGAN	high	972	560	0.388

Table 5.3: Global codebook distribution statistics under global perturbations. Top- m mass is the fraction of all token assignments explained by the m most frequent codes at that noise level.

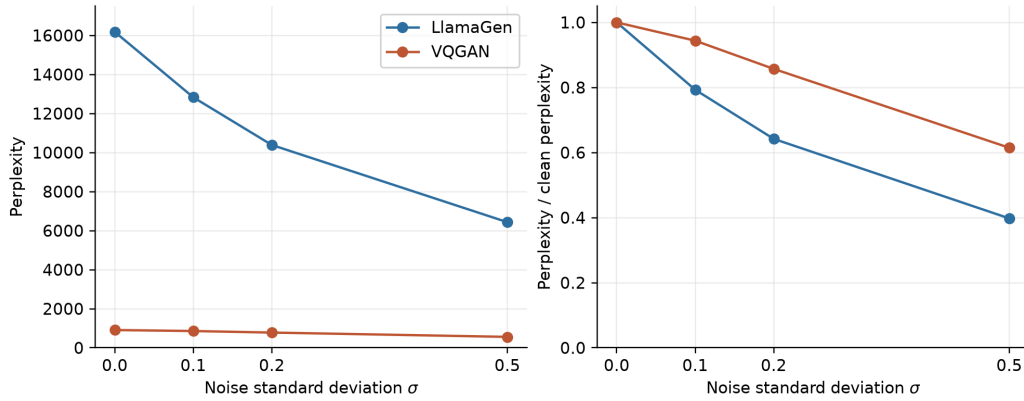


Figure 5.6: Effective codebook perplexity under global perturbations, shown both in absolute units and relative to each checkpoint’s clean baseline. Both empirical vocabularies contract as noise increases.

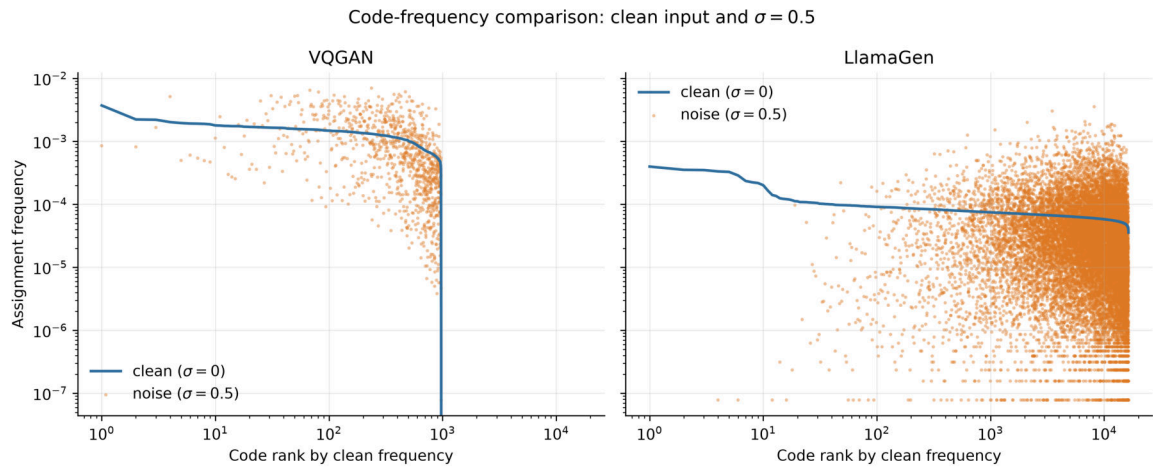


Figure 5.7: Per-code assignment frequencies for clean inputs and $\sigma = 0.5$. Within each tokenizer, codes are ordered once by their clean frequency; the noisy points retain that fixed code ordering.

6 | Local Perturbations in Image Space

Global perturbations mix the response associated with one token’s corresponding image region with responses caused by changes elsewhere. This chapter perturbs one image patch and measures how far assignment changes spread beyond it. Chapter 5 showed assignment instability under global noise; the present experiment tests whether changes also appear outside a bounded patch.

Finite convolutional receptive fields create overlapping local dependencies, while attention and spatial normalization can couple distant positions. The two evaluated encoders use VQGAN-style convolutional backbones with 3×3 stride-2 downsampling, 3×3 residual convolutions, GroupNorm, and attention at the lowest-resolution level. These facts identify candidate mechanisms, not the cause of the measured response. EdiBERT previously observed token changes outside a local image mask on a VQGAN model [16]; here the distance profile measures their frequency across the ImageNet validation set for both checkpoints.

6.1. Experimental Setup

6.1.1. Patch-Noise Protocol

Let $x \in [-1, 1]^{H \times W \times 3}$ be an input image and let P denote a square patch. The binary mask M_P equals one inside the patch and zero outside it. For a noise level σ , the perturbed image is

$$x_{\sigma, P} = \text{clip}(x + \sigma \epsilon \odot M_P, -1, 1), \quad \epsilon \sim \mathcal{N}(0, I). \quad (6.1)$$

Only pixels inside P receive additive noise. The clipping operation keeps the image in the valid pixel range.

Both tokenizers produce a 16×16 token grid. The analysis uses an 8×8 token-space patch, corresponding to a 128×128 pixel region. Its top-left corner is sampled uniformly

from valid grid locations for each of the 50,000 ImageNet validation images. The reported levels are $\sigma \in \{0.1, 0.25, 0.5\}$. The two runs use the same images and nominal settings but different random patch locations and noise realizations. Strictly paired model comparisons would require shared perturbations and a rerun.

The masked variant fills the selected patch with black in the internal image range. With fill value $c = -1$, it is defined as

$$x_{\text{mask},P} = (1 - M_P) \odot x + M_P \odot c. \quad (6.2)$$

6.1.2. Inside and Outside Response

Let $z_0 \in \{1, \dots, K\}^{16 \times 16}$ be the clean token grid and $z_{\sigma,P}$ be the token grid obtained from the patch-perturbed image. Let I_P be the set of token positions covered by the patch and let O_P be the complement. We define the inside response as the fraction of token assignments within the selected patch that change:

$$R_{\text{in}}(\sigma, P) = \frac{1}{|I_P|} \sum_{i \in I_P} \mathbf{1}[z_{\sigma,P,i} \neq z_{0,i}]. \quad (6.3)$$

We define the outside response as the fraction of token assignments outside the selected patch that change:

$$R_{\text{out}}(\sigma, P) = \frac{1}{|O_P|} \sum_{i \in O_P} \mathbf{1}[z_{\sigma,P,i} \neq z_{0,i}]. \quad (6.4)$$

We define the encoder leakage ratio as

$$\lambda_{\text{enc}}(\sigma, P) = \frac{R_{\text{out}}(\sigma, P)}{R_{\text{in}}(\sigma, P) + \varepsilon} \quad (6.5)$$

where $\varepsilon = 10^{-8}$ prevents division by zero. The reported λ_{enc} is the mean of per-image ratios, not the ratio of the tabulated mean responses. All responses are fractions of token assignments that differ from the clean encoding. A value of λ_{enc} near zero indicates that changes are mostly confined to the patch, while a value near one indicates similar change rates inside and outside it. We retain R_{in} and R_{out} alongside the ratio because the same ratio can arise from different absolute responses.

6.1.3. Distance Profile

The inside and outside responses do not show how far token changes spread. Each token position i is therefore assigned a Manhattan grid distance $d(i, P)$ to the nearest token in I_P : distance zero is inside the patch, distance one is the first ring of neighboring token cells, and larger values move farther across the token grid. For image n , let $A_{n,d}$ contain its token positions at distance d from its patch P_n . The reported profile pools token counts across images:

$$R_d(\sigma) = \frac{\sum_n \sum_{i \in A_{n,d}} \mathbf{1}[z_{\sigma, P_n, i}^{(n)} \neq z_{0, i}^{(n)}]}{\sum_n |A_{n,d}|}. \quad (6.6)$$

Each eligible token has equal weight, so images with more tokens in a band contribute more weight. An image with no tokens at distance d contributes neither flips nor observations there; a globally empty band is undefined, rather than assigned zero. The results report the full profile and three summaries: inside tokens ($d = 0$), boundary-neighbor tokens ($d = 1$), and far-field tokens ($d \geq 6$). Band summaries use the same pooled counts. Appendix C retains the per-distance probabilities and their token denominators.

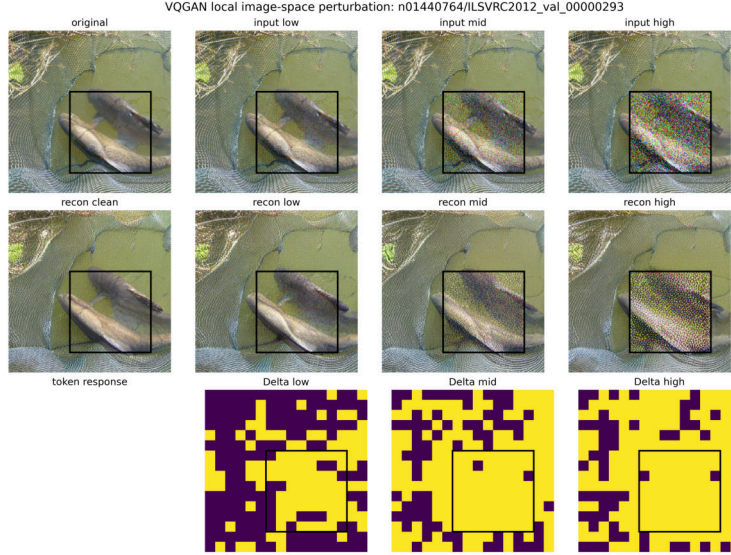
All 50,000 images contribute at distances zero through eight. Beyond that, eligibility depends on patch placement. At $d = 16$, only corner placements contribute, with one opposite-corner token each: the saved denominator is 2,488 tokens from 2,488 images. Tail values therefore describe eligible patch and image positions, not a fixed image population followed over distance.

6.2. Experimental Results

Qualitative patch perturbations. Figure 6.1 shows selected examples for the two tokenizers. Each panel contains the clean image, the patch-noisy inputs, the corresponding reconstructions, and binary token-response maps. The black rectangle marks the perturbed patch in pixel space and the corresponding token block in the response map. Highlighted cells outside the rectangle are token-assignment changes outside the patch.

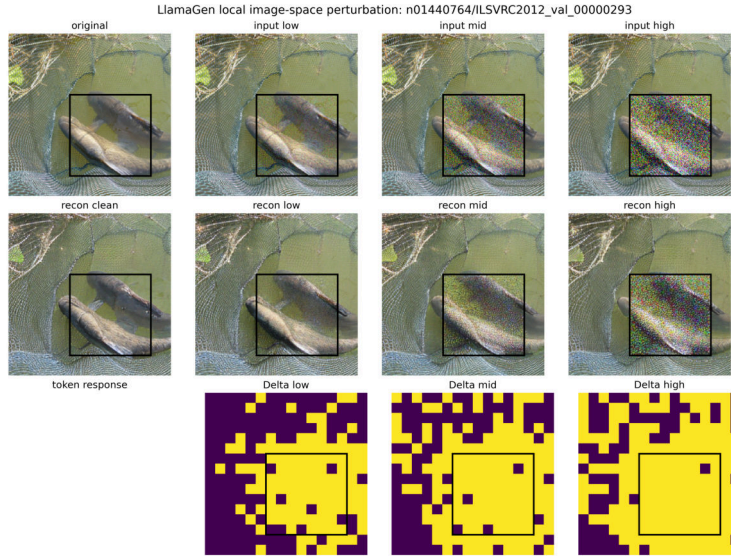
Local response and leakage. Table 6.1 reports the mean inside response R_{in} , outside response R_{out} , encoder leakage ratio λ_{enc} , and far-field response. A high inside response is expected because the corresponding image pixels are directly perturbed. At $\sigma = 0.1$, VQGAN changes 78.8% of inside assignments and LlamaGen changes 82.2%. At $\sigma = 0.5$, the inside response is almost saturated for both.

(a) VQGAN



Sample ILSVRC2012_val_00000293: $\sigma = 0.1$ ($R_{\text{in}} = 0.766$, $R_{\text{out}} = 0.297$, $\lambda_{\text{enc}} = 0.388$); $\sigma = 0.25$ (0.969, 0.630, 0.651); $\sigma = 0.5$ (0.969, 0.724, 0.747).

(b) LlamaGen



Sample ILSVRC2012_val_00000293: $\sigma = 0.1$ ($R_{\text{in}} = 0.875$, $R_{\text{out}} = 0.323$, $\lambda_{\text{enc}} = 0.369$); $\sigma = 0.25$ (0.938, 0.526, 0.561); $\sigma = 0.5$ (0.984, 0.656, 0.667).

Figure 6.1: Selected local patch-noise examples for (a) VQGAN and (b) LlamaGen using the same input ID. The binary response maps use dark cells for unchanged assignments (0) and yellow cells for changed assignments (1); the marked rectangle identifies the perturbed patch. Metrics below each panel are sample-specific and ordered as R_{in} , R_{out} , and λ_{enc} . This example has higher leakage than the dataset mean at the two stronger levels and is not representative of the average response.

Tokenizer	σ	R_{in}	R_{out}	λ_{enc}	Far field
VQGAN	0.10	0.788	0.299	0.375	0.174
VQGAN	0.25	0.952	0.441	0.462	0.265
VQGAN	0.50	0.988	0.521	0.527	0.333
LlamaGen	0.10	0.822	0.302	0.361	0.178
LlamaGen	0.25	0.960	0.428	0.444	0.260
LlamaGen	0.50	0.991	0.505	0.509	0.328

Table 6.1: Inside and outside token response under local patch-noise perturbations. The far-field column reports the mean token flip rate for positions at distance at least six token cells from the perturbed patch.

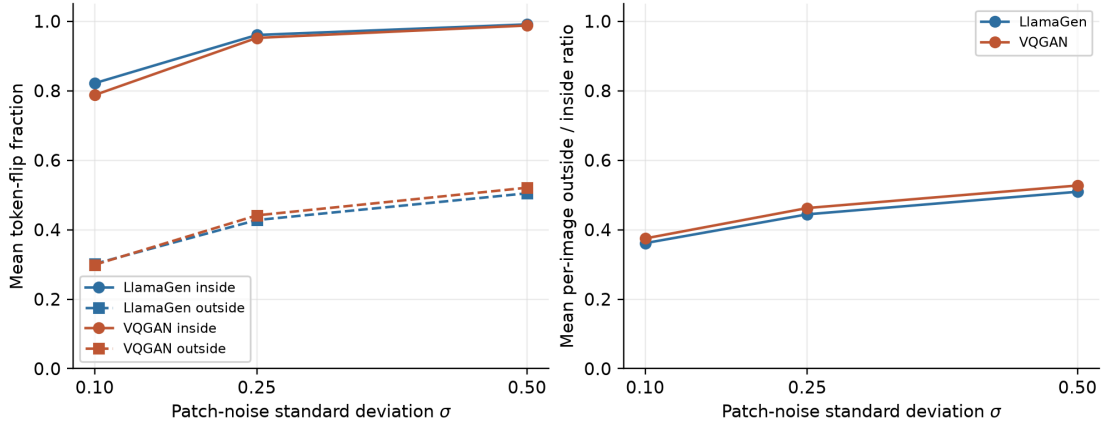


Figure 6.2: Mean token response inside and outside the perturbed patch, with the corresponding leakage ratio. Outside-token changes increase monotonically with noise level for both tokenizers, and the outside/inside ratio also increases.

Token assignments also change outside the patch at the lowest tested level: R_{out} is 29.9% for VQGAN and 30.2% for LlamaGen at $\sigma = 0.1$. It rises to 44.1% and 42.8%, respectively, at $\sigma = 0.25$, then to 52.1% and 50.5% at $\sigma = 0.5$. At the largest tested value, outside assignments change about half as often as inside assignments. A lower-noise sweep would be needed to estimate when R_{out} approaches zero.

Distance from the perturbed patch. Figure 6.3 plots the response to the image perturbation as a function of distance from the patch. It declines sharply over the first few token cells but remains nonzero farther away. Table 6.2 summarizes the curves using descriptive distance bands. The chosen far-field band, $d \geq 6$, separates distant positions from the immediate rings; it is not a measured boundary between local and global mechanisms. In this band, 17.4% of VQGAN assignments and 17.8% of LlamaGen assignments change at $\sigma = 0.1$. At $\sigma = 0.5$, the fraction is roughly one third.

Tokenizer	σ	Inside	Boundary	Far field
VQGAN	0.10	0.788	0.597	0.174
VQGAN	0.25	0.952	0.827	0.265
VQGAN	0.50	0.988	0.917	0.333
LlamaGen	0.10	0.822	0.613	0.178
LlamaGen	0.25	0.960	0.832	0.260
LlamaGen	0.50	0.991	0.922	0.328

Table 6.2: Distance-band summary of token response. Inside denotes $d = 0$, boundary denotes $d = 1$, and far field denotes positions with Manhattan distance $d \geq 6$ from the perturbed patch.

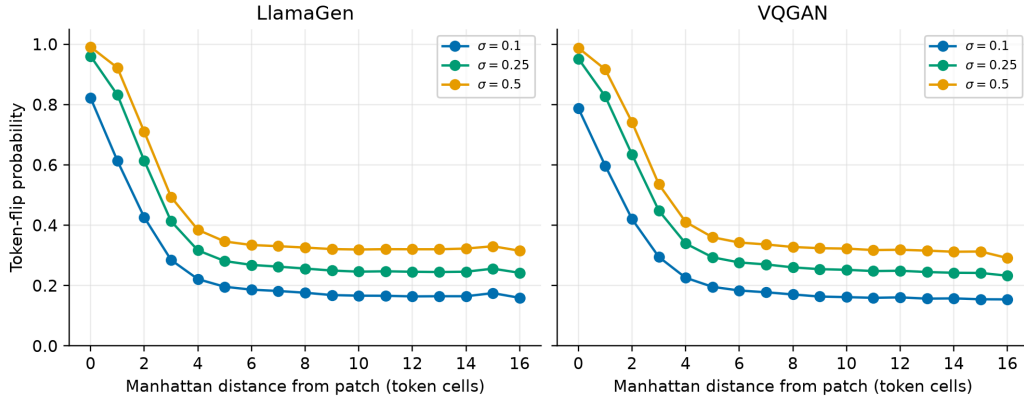


Figure 6.3: Pooled token flip probability by distance from the perturbed patch. The response is strongest inside and near the patch, but remains measurable far from it. Beyond distance eight, the contributing image population depends on patch placement; Table C.3 reports token denominators.

Masked patch intervention. The masked intervention defined in Equation 6.2 replaces the same local patch with black. Table 6.3 shows that this nearly saturates the inside response for both tokenizers, while the response to the image perturbation outside the patch is 56.3% for VQGAN and 49.5% for LlamaGen. Thus, a change confined to one quarter of the image changes token assignments across roughly half of the remaining grid.

6.3. Discussion

Both tokenizers show the same measured pattern: token changes are most frequent inside and near the perturbed patch, decline over the nearby token cells, and remain nonzero at distant positions. LlamaGen has a slightly higher inside response at $\sigma = 0.1$ and a slightly lower leakage ratio at each tested value, but the spatial pattern is shared. These small cross-checkpoint differences are descriptive because the patch locations and noise

Tokenizer	R_{in}	R_{out}	λ_{enc}
VQGAN	0.998	0.563	0.564
LlamaGen	1.000	0.495	0.495

Table 6.3: Inside and outside token response for the masked-patch intervention. The mask is a stronger local corruption than Gaussian patch noise and produces a correspondingly high outside response.

draws were not matched.

At $\sigma = 0.1$, inside-patch flip rates exceed the global-noise rates from Chapter 5. The input context and spatial support of the noise differ, so equal nominal σ does not make these the same intervention.

The evaluated architecture contains overlapping 3×3 convolutions, GroupNorm, and an attention layer. A small feature movement can also cross a nearest-neighbor quantization boundary and change the assignment at that position, although pointwise quantization does not transmit changes between positions. The measurements do not isolate the contribution of these components. Future work can investigate attention, spatial normalization, and receptive-field overlap through ablations and intermediate-feature analysis.

This result complements Chapter 5. Global noise changed many token assignments when every image region was perturbed. The study of perturbation locality shows how far those changes spread when the image modification is bounded: token assignments change well beyond the selected patch. This establishes non-local dependence of code identity, but the experiment does not decode only the outside flips to measure their separate image contribution.

7 | Local Perturbations in Token Space

The previous chapter showed that a local image-space perturbation causes token changes well beyond the corrupted patch. This chapter examines the complementary decoder-side question: when token IDs inside a bounded region of the latent grid are replaced directly, how far does the decoded image change?

The experiment replaces token IDs inside a contiguous rectangular region and measures the decoded change inside and outside the corresponding image region. It asks how severe the change is and how far it extends beyond the edited patch.

A discrete image tokenizer represents an image as a spatial grid of codebook indices. Editing a bounded region of this grid is expected to affect mainly the corresponding region of the decoded image. Decoder locality matters because autoregressive image models may make local token errors: if the decoder spreads such errors widely, a local sequence error can become a broader image artifact.

The replacement strategies test whether nearby codebook vectors decode to similar image content and whether more distant vectors cause larger changes. Qiu et al. use nearest-neighbor token replacement as a training perturbation for sampling errors [25]. Here the checkpoints remain frozen, and the question is the spatial response of their decoders rather than whether robustness training improves generation. Nearest-neighbor replacement tests a small change in the embedding space; farthest replacement tests a large one. Orthogonal replacement selects a vector with cosine similarity near zero, and random replacement samples a codebook ID without conditioning on its distance from the original vector. The resulting distances still depend on codebook geometry. We expect these strategies to produce different levels of change, but their decoded effects must be measured.

7.1. Experimental Setup

7.1.1. Token-Edit Protocol

Let $z_0 \in \{1, \dots, K\}^{h \times w}$ denote the clean token grid produced for an image, and let B_{tok} be a contiguous rectangular region of that grid. For an edit mode m , the edited token grid $z^{(m)}$ is

$$z_{i,j}^{(m)} = \begin{cases} T_m(z_{0,i,j}), & (i, j) \in B_{\text{tok}}, \\ z_{0,i,j}, & (i, j) \notin B_{\text{tok}}, \end{cases} \quad (7.1)$$

where T_m gives a replacement token. For the random mode it denotes a fresh draw at each edited position, rather than a fixed mapping of code IDs. The experiment uses four mappings:

- **Closest**: the nearest alternative codebook vector;
- **Orthogonal**: a codebook vector with cosine similarity close to zero;
- **Random**: a uniformly sampled codebook vector;
- **Farthest**: the most distant codebook vector.

For VQGAN, the closest, orthogonal, and farthest mappings are restricted to the empirically active code subset identified in Chapter 4. VQGAN uses only part of its codebook on the reference dataset. Uniform-random replacement is different: it samples from the full codebook for both models. For VQGAN this means that random replacement may introduce inactive codes, so the random edit tests both replacement and the use of previously inactive codes.

Closest and farthest replacement minimize and maximize squared Euclidean distance within the candidate set, excluding the original code ID. The precomputation uses raw VQGAN embedding vectors and unit-normalized LlamaGen vectors. Orthogonal replacement minimizes absolute cosine similarity, also excluding the original ID; normalization uses a 10^{-12} stabilizer in the vector norm. It does not sample from a tolerance band around zero. The three geometric rules produce fixed lookup tables reused at every occurrence of a code. Ties select the first candidate in increasing code-ID order. Random replacement instead draws independently for each edited occurrence from all K IDs, with replacement and without excluding the original ID. Draws are generated afresh for each patch fraction, so a selected position can retain its original token with probability $1/K$.

The clean and edited reconstructions are

$$x_{\text{clean}} = D(z_0), \quad x_{\text{edit}}^{(m)} = D(z^{(m)}), \quad (7.2)$$

where D is the tokenizer decoder. All image changes are measured against x_{clean} , so ordinary reconstruction error relative to the input does not enter the perturbation measurement.

The main comparisons use a patch covering 25% of the token grid. A patch-size sweep also tests target fractions of 10%, 50%, and 75%. The main analysis uses ImageNet validation; ImageNet-V2 is used to test whether the observed trends remain stable under a natural distribution shift.

Images are resized and center-cropped to 256×256 using the model-specific preprocessing described in Appendix B. Decoder outputs are converted from the models' internal $[-1, 1]$ range to $[0, 1]$ before metrics are computed. Patch locations are reproducible within each run. The two tokenizers did not receive shared random replacement draws, so comparisons describe aggregate behavior.

Both tokenizers produce a 16×16 grid. For each image, a square patch is sampled at a random valid position for the largest target fraction. The smaller patches are centered within that patch using integer offsets $\lfloor (14 - s)/2 \rfloor$ on both axes for side length s . Thus the offsets for sides 5, 8, 11, and 14 are 4, 3, 1, and 0 tokens; an odd margin leaves the extra token on the bottom and right. The largest patch's row and column origins are sampled uniformly from $\{0, 1, 2\}$ and reused across edit modes. The square side lengths are 5, 8, 11, and 14 tokens, corresponding to actual grid fractions of 9.8%, 25.0%, 47.3%, and 76.6%, respectively.

7.1.2. Fidelity Metrics

Decoder locality alone does not describe the severity of an edit. A low outside response may coincide with either a negligible local change or severe damage confined to the patch. Using the definitions and metric references in Chapter 3, the experiment measures PSNR, SSIM, and LPIPS between x_{clean} and $x_{\text{edit}}^{(m)}$, both over the full image and within B_{px} . Patch-level metrics quantify edit severity; the inside/outside response and leakage ratio quantify spatial propagation. The comparison is made against the clean reconstruction rather than the original image so that the measurement isolates the decoder response to the token edit, not the tokenizer's ordinary reconstruction error.

7.1.3. Decoder-Locality Metrics

To measure how far decoded changes spread, we compare the average response inside and outside the edited region. The per-pixel measurement follows the pixel-error metrics

introduced in Chapter 3; the channel-averaged RMSE used here is defined below. The decoder response is measured relative to the clean reconstruction. For pixel position (u, v) , the channel-averaged root-mean-square change is

$$\Delta^{(m)}(u, v) = \sqrt{\frac{1}{3} \sum_{c=1}^3 \left(x_{\text{edit}}^{(m)}(u, v, c) - x_{\text{clean}}(u, v, c) \right)^2}. \quad (7.3)$$

Let B_{px} denote the pixel-space region corresponding to the edited token patch. Mean inside and outside responses are

$$\Delta_{\text{in}}^{(m)} = \frac{1}{|B_{\text{px}}|} \sum_{(u,v) \in B_{\text{px}}} \Delta^{(m)}(u, v), \quad \Delta_{\text{out}}^{(m)} = \frac{1}{|\overline{B}_{\text{px}}|} \sum_{(u,v) \notin B_{\text{px}}} \Delta^{(m)}(u, v). \quad (7.4)$$

Here $\overline{B}_{\text{px}}$ is the complement of B_{px} within the image. Its cardinality in the denominator is the number of *outside* pixels, matching the summation over $(u, v) \notin B_{\text{px}}$. The decoder leakage ratio is

$$\lambda_{\text{dec}}^{(m)} = \frac{\Delta_{\text{out}}^{(m)}}{\Delta_{\text{in}}^{(m)} + \varepsilon}. \quad (7.5)$$

A small positive $\varepsilon = 10^{-8}$ stabilizes the denominator. A low ratio means that the average pixel change is smaller outside than inside the edited region. Δ_{in} measures local edit strength, while Δ_{out} measures leakage outside the intended image region. The analysis also reports the 99th and 99.9th percentiles of outside-patch change to capture strong but spatially sparse leakage.

The leakage ratio is calculated per image. Its distribution can become unstable when $\Delta_{\text{in}}^{(m)}$ is close to zero: a small outside response divided by an even smaller inside response can yield a large ratio. The median is therefore the primary summary of proportional leakage; means and standard deviations are retained in the archived per-image tables. The absolute inside and outside responses are reported alongside $\lambda_{\text{dec}}^{(m)}$.

7.2. Experimental Results

7.2.1. Qualitative Decoder Response

Figure 7.1 shows one ImageNet validation example for a 25% token patch. It shows the clean reconstruction, the four edited reconstructions, and the corresponding pixel-change maps for both tokenizers. The nearest-neighbor edit preserves the fish and surrounding net with only modest local changes. Orthogonal and random replacement alter the

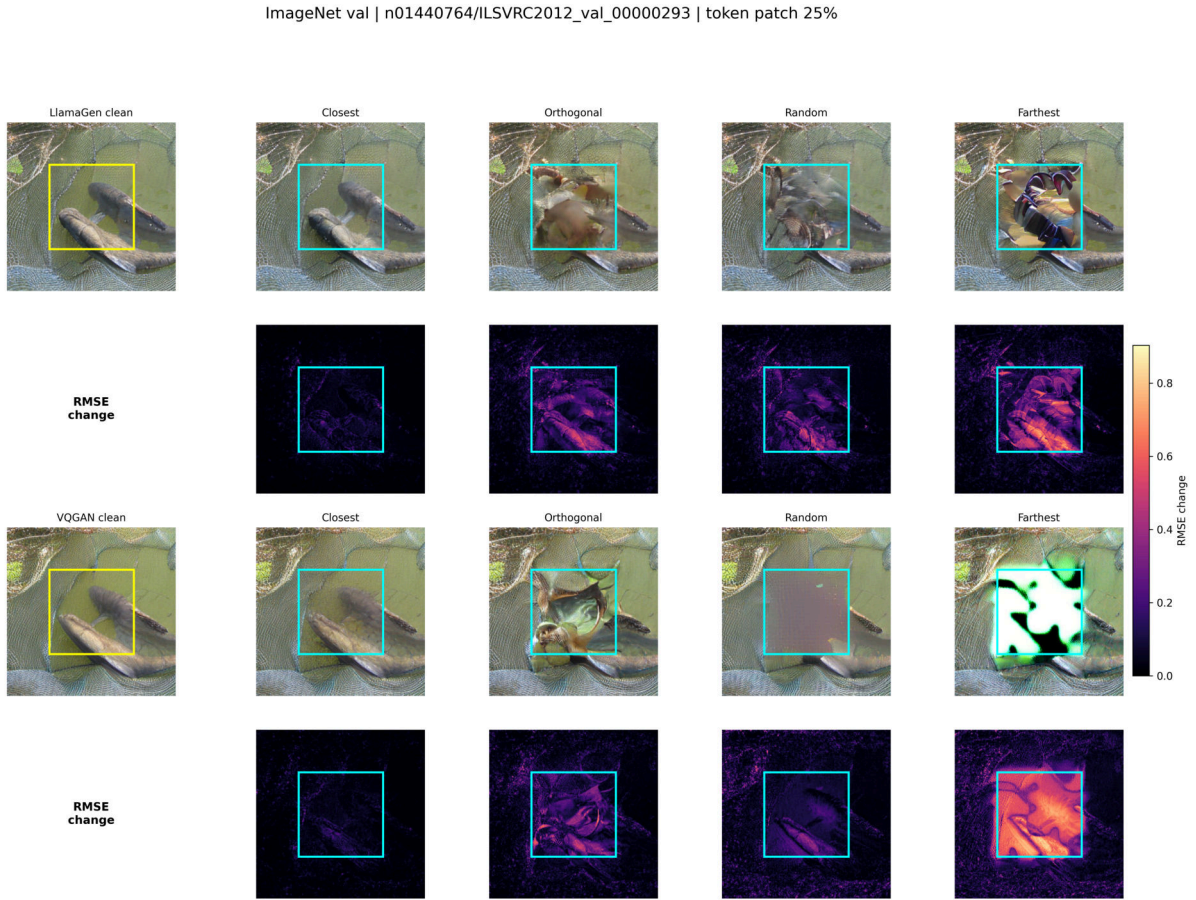


Figure 7.1: Qualitative decoder response to a 25% token-patch perturbation. Yellow marks the region in each clean reconstruction; cyan marks the same region in edited reconstructions and RMSE heatmaps. RMSE heatmaps show pixel change relative to each tokenizer’s clean reconstruction; the figure is illustrative, with no recorded representative-sampling rule. Its shared linear color scale can hide small outside changes; the numerical summaries and distance profile establish those changes.

content inside the patch more visibly, while the farthest edit produces the strongest local transformation for both models. For VQGAN, the farthest edit becomes an extreme high-contrast artifact, consistent with the large quantitative response reported below.

7.2.2. Edit Severity

Table 7.1 and Figure 7.2 compare the four replacement strategies at the 25% patch level. Closest replacement is the mildest intervention for both tokenizers. Its patch PSNR is 20.36 dB for LlamaGen and 20.00 dB for VQGAN, while patch LPIPS remains at 0.134 and 0.182. These edits preserve recognizable content in Figure 7.1; the more distant replacements visibly change the patch. PSNR measures pixel error, whereas LPIPS mea-

Tokenizer	Edit mode	Δ_{in}	Patch PSNR	Patch LPIPS
LlamaGen	Closest	0.067	20.36	0.134
LlamaGen	Orthogonal	0.222	11.45	0.599
LlamaGen	Random	0.220	11.48	0.592
LlamaGen	Farthest	0.356	7.74	0.702
VQGAN	Closest	0.072	20.00	0.182
VQGAN	Orthogonal	0.239	10.66	0.614
VQGAN	Random	0.200	12.83	0.749
VQGAN	Farthest	0.607	3.74	0.790

Table 7.1: Edit severity for a 25% token patch on ImageNet validation. Values are means over 50,000 images. The inside RMSE is repeated here to relate pixel-response magnitude to patch fidelity; the locality table also reports the corresponding outside response.

sures differences in learned image features. Moving away from the nearest code sharply increases perceptual change. Orthogonal replacement raises patch LPIPS to 0.599 for LlamaGen and 0.614 for VQGAN. The random edit should be interpreted separately from the three geometric edits because, for VQGAN, it may sample inactive codes from the full nominal codebook.

For LlamaGen, orthogonal and random replacement are almost indistinguishable: their patch LPIPS values are 0.599 and 0.592, and their inside responses are 0.222 and 0.220. VQGAN behaves differently. Random replacement has a lower mean inside RMSE than orthogonal replacement, 0.200 versus 0.239, but a much higher patch LPIPS, 0.749 versus 0.614. The disagreement between RMSE and LPIPS means that average pixel displacement does not account for the perceptual effect of the VQGAN random edit.

Farthest replacement is the strongest intervention. At the 25% level, LlamaGen reaches a patch LPIPS of 0.702 and a patch PSNR of 7.74 dB. VQGAN is more severely affected under its own farthest-code edit, with patch LPIPS 0.790, patch PSNR 3.74 dB, and an inside response of 0.607. Farthest-code edits are defined separately in the two codebooks, so this is not a severity-matched comparison of the decoders.

Figure 7.3 shows that patch-level perceptual change generally increases with patch fraction. Closest replacement stays well below the other interventions across the sweep. For LlamaGen, orthogonal and random edits continue to track each other closely, rising from approximately 0.51–0.52 at the 10% level to approximately 0.67–0.68 at 75%. Farthest replacement remains highest, increasing from 0.630 to 0.782.

VQGAN shows a wider separation between edit modes. Its closest-edit LPIPS increases from 0.139 to 0.230. Random replacement rises from 0.598 to 0.879, and overtakes farthest replacement at the 75% level, where farthest LPIPS is 0.861. The farthest edit still has

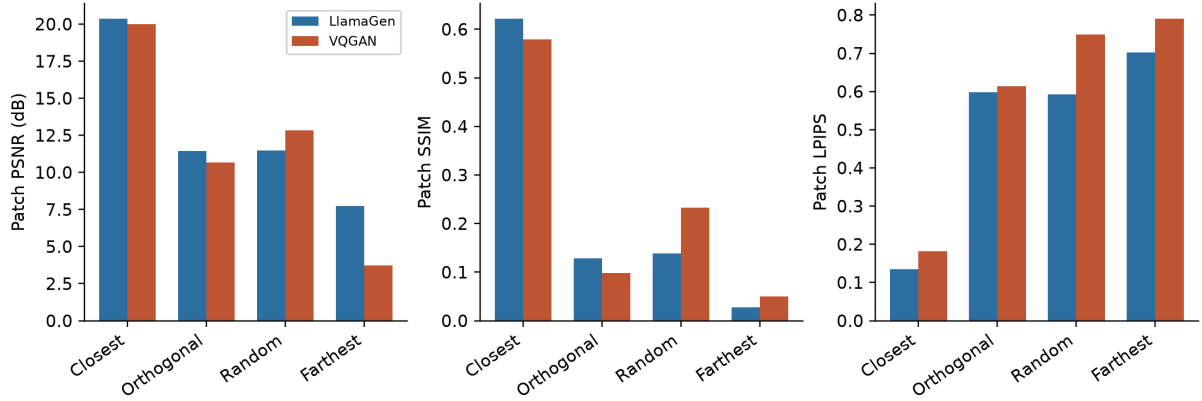


Figure 7.2: Patch-level PSNR, SSIM, and LPIPS for the four edit modes at a 25% token-patch fraction.

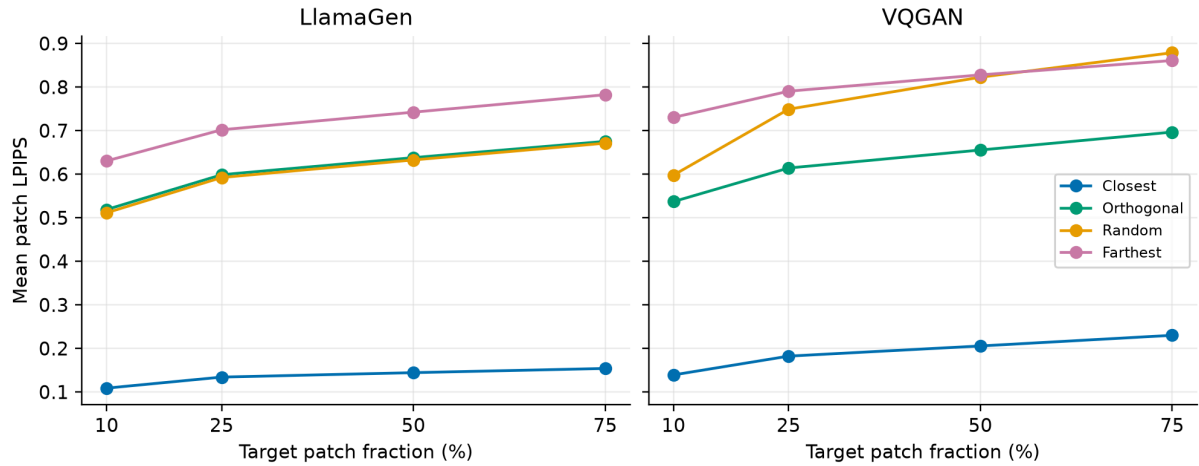


Figure 7.3: Patch-level perceptual change as the edited token-patch fraction increases.

by far the largest inside RMSE response. The two metrics rank the strongest VQGAN edit differently.

7.2.3. Inside and Outside Decoder Response

Table 7.2 separates the response inside the edited region from the response outside it. At the 25% patch level, all edit modes produce a substantially larger mean change inside the patch. For LlamaGen, outside response ranges from 0.013 for closest replacement to 0.050 for farthest replacement. For VQGAN, it ranges from 0.019 to 0.090. The decoder response is spatially concentrated, but not confined to the edited region.

The heatmaps in Figure 7.1 show the same pattern: the response does not stop exactly at the cyan boundary. Most change remains within or near the edited region, but

Tokenizer	Edit mode	Δ_{in}	Δ_{out}	$\tilde{\lambda}_{\text{dec}}$
LlamaGen	Closest	0.067	0.013	0.193
LlamaGen	Orthogonal	0.222	0.039	0.176
LlamaGen	Random	0.220	0.038	0.175
LlamaGen	Farthest	0.356	0.050	0.141
VQGAN	Closest	0.072	0.019	0.263
VQGAN	Orthogonal	0.239	0.046	0.187
VQGAN	Random	0.200	0.050	0.249
VQGAN	Farthest	0.607	0.090	0.147

Table 7.2: Mean inside and outside response and median decoder leakage ratio for a 25% token patch on ImageNet validation.

lower-amplitude changes extend into the surrounding image, including under closest replacement.

Even nearest-neighbor replacement changes pixels outside the patch, as the heatmaps in Figure 7.1 also show. Farthest replacement causes the greatest absolute change both inside and outside. Its small ratio therefore reflects severe inside damage and should not be read as evidence that the replacement is harmless outside the patch.

The outside response also contains localized high-amplitude changes that are not visible from its mean alone. At the 25% patch level, VQGAN farthest replacement has a mean outside response of 0.090, but its mean outside 99th and 99.9th percentiles reach 0.673 and 0.854. For LlamaGen, the corresponding values are 0.050, 0.328, and 0.554. Closest replacement has much smaller outside tails, with 99th-percentile values of 0.105 for LlamaGen and 0.148 for VQGAN. These tail statistics matter because mean outside response can hide localized artifacts: strong changes can occur at a small subset of outside pixels even when the average outside response remains low. Table C.13 records the quoted farthest-edit tails.

Figure 7.4 shows how these quantities change with patch size. The mean inside response is comparatively stable for closest, orthogonal, and farthest replacement. For example, VQGAN’s farthest-edit inside response changes only from 0.592 at the 10% level to 0.613 at 75%. LlamaGen’s corresponding response rises from 0.329 to 0.410. By contrast, the outside response increases with patch size. For VQGAN farthest replacement, the outside response rises from 0.053 to 0.213; for LlamaGen it rises from 0.030 to 0.120. The outside region also changes geometry: an edit covering roughly three-quarters of a 16×16 grid leaves only a narrow outside strip. Thus, a higher outside mean at larger patch fractions does not by itself show that an individual edit spreads farther from its boundary.

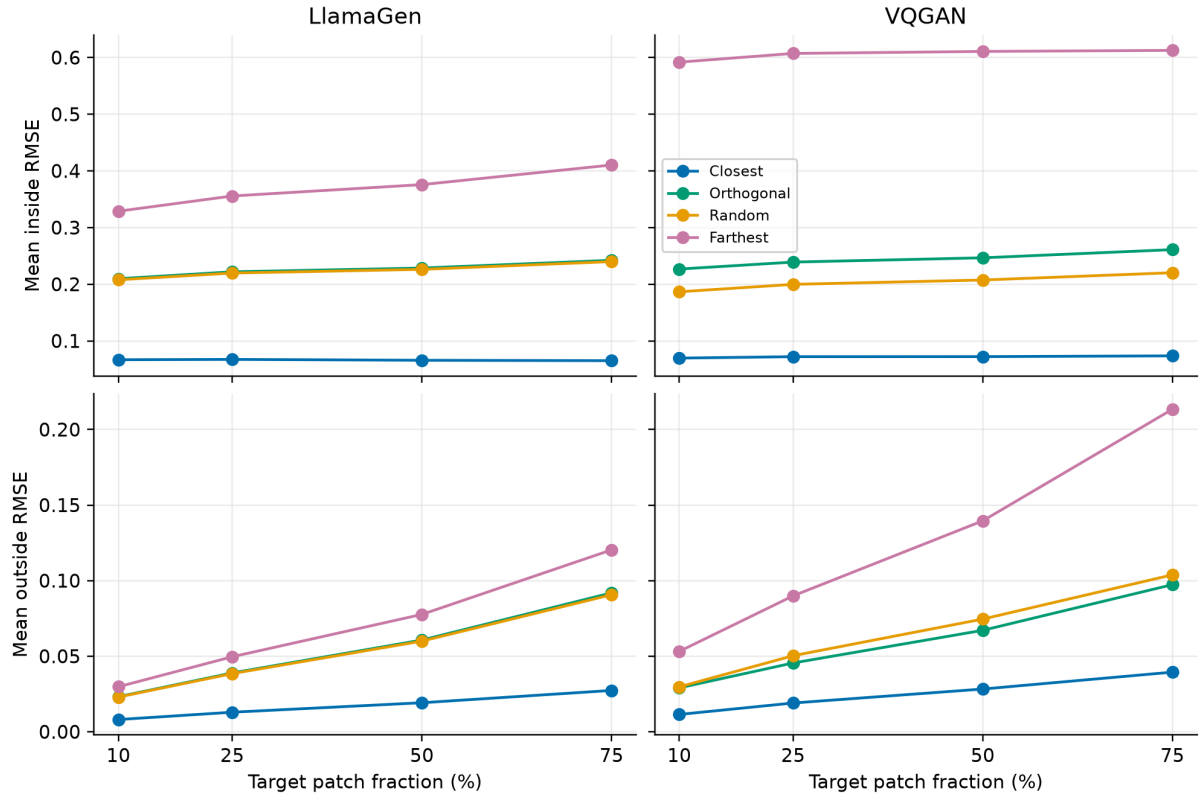


Figure 7.4: Mean image-space response inside and outside the edited token patch as the patch fraction increases.

Distance from the edited boundary. To measure spatial decay directly, we revisited saved clean and edited reconstruction PNGs for the 25% patch setting on the same fixed-seed 500 ImageNet IDs for both models. For each edit mode, we averaged per-pixel RGB RMSE over outside pixels in 16-pixel-wide Euclidean distance bands from the patch boundary, then averaged those band values across images. Figure 7.6 shows a steep decline near the boundary but no abrupt cutoff. For random replacement, LlamaGen’s mean response falls from 0.091 in the first 16-pixel band to 0.016 at 80–96 pixels; VQGAN’s falls from 0.106 to 0.031. Farthest replacement retains the largest far-field response (0.021 for LlamaGen and 0.043 for VQGAN in the last band). These are descriptive values from eight-bit outputs, not an estimate of a fixed receptive-field radius; the models’ replacement codes are not matched in severity.

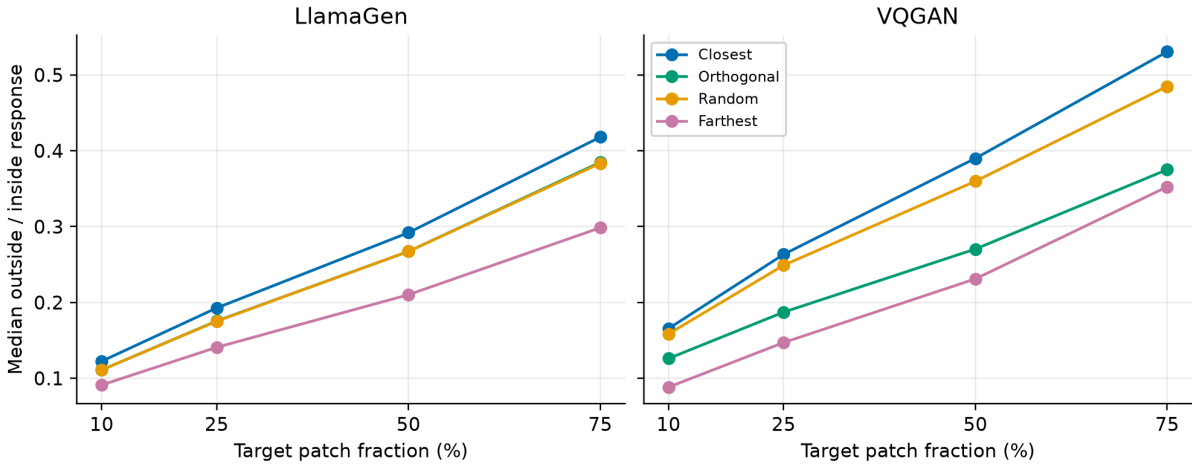


Figure 7.5: Median decoder leakage ratio as a function of token-patch fraction and edit mode.

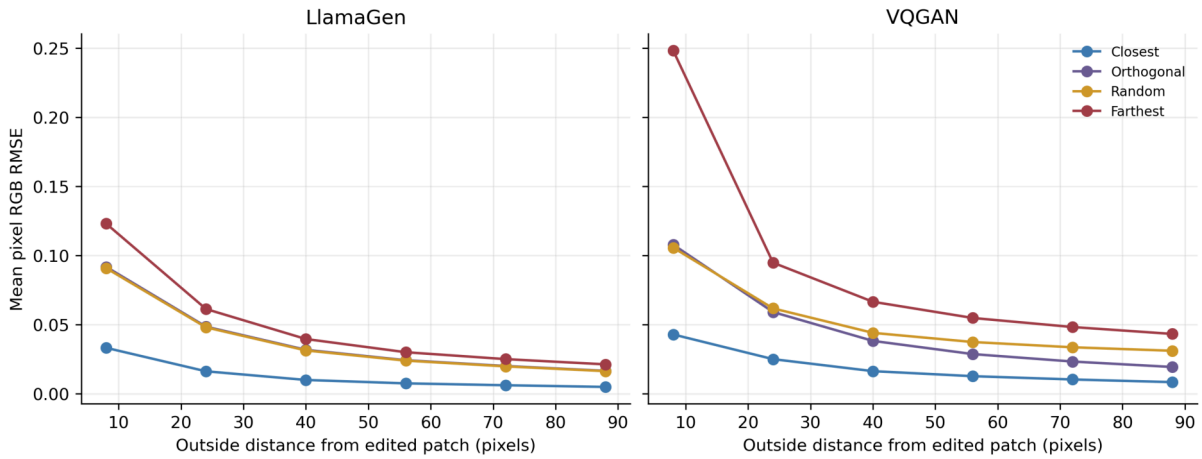


Figure 7.6: Mean pixel-level RGB RMSE outside a 25% token edit, binned by Euclidean distance to the edited patch boundary. Each curve averages the same 500 ImageNet validation IDs per checkpoint; clean and edited PNGs are eight-bit saved outputs. The response declines with distance but remains measurable in the farthest outside band.

The patch-size trend also appears in Figure 7.5. At the 75% level, median leakage ratios range from 0.299 to 0.418 for LlamaGen and from 0.352 to 0.531 for VQGAN. The closest and random VQGAN edits have the highest proportional leakage at large patch sizes. The plots use median per-image ratios throughout; near-zero inside responses can distort the mean.

7.2.4. Robustness under Dataset Shift

We test ImageNet-V2 to check whether the measured decoder locality persists beyond the in-distribution reference images. Patch-LPIPS differences from the ImageNet validation baseline are small across both models, all four edit modes, and all patch fractions: the largest absolute difference is 0.0068. These differences are small relative to the contrasts between edit modes. Their practical importance has not been established; this comparison does not provide an uncertainty estimate or a test of equivalence between datasets.

The other metrics are similarly stable. Across the full comparison, the largest absolute change in mean outside response is 0.0016. At the 25% patch level, leakage-ratio differences range from -0.0030 to -0.0080 , and patch-PSNR differences remain within 0.17 dB. These differences are small relative to those among edit modes and tokenizers. ImageNet-V2 preserves the ordering and approximate magnitude found on ImageNet validation.

7.3. Discussion

A local token-space perturbation produces a spatially concentrated decoder response, but the response is not strictly local. Mean change is always stronger inside the corresponding image region than outside it, and the qualitative heatmaps preserve a clear spatial relationship to the edited patch. Outside response is measurable even for nearest-neighbor replacement and grows as the edited region becomes larger. Farthest-code replacement causes large absolute changes both inside and outside the patch.

Orthogonal and random replacements are nearly equivalent for LlamaGen. For VQGAN, random replacement is more disruptive under LPIPS than the orthogonal edit despite a smaller average pixel response.

VQGAN random replacement can introduce codes absent from its clean ImageNet assignments. Its response therefore combines code distance with exposure to codes unused on those inputs.

The same within-checkpoint edit patterns persist across patch fractions. At the 25% level, the VQGAN farthest-code edit has an inside response of 0.607, compared with 0.356 for LlamaGen’s separately defined farthest-code edit; its patch PSNR is approximately 4 dB lower. These edits are not severity-matched across codebooks, so the difference does not isolate decoder sensitivity. Neither decoder confines the effect completely.

Chapter 6 measures token identity after image edits; this chapter measures pixel change after token edits. Both detect responses outside the edited region, but their different inter-

ventions and units do not support a ranking of encoder versus decoder locality. Complete condition summaries and across-image variability are in Appendix C.

ImageNet-V2 preserves the same ordering and approximate response magnitude. Decoder locality therefore persists under this mild shift to natural images from different sources. Testing whether it persists under stronger style and domain changes remains future work.

8 | Robustness to Distribution Shift

The preceding experiments measured responses to controlled noise and local perturbations on in-distribution ImageNet images or their token grids. Inputs encountered outside training can differ systematically in style, content, and layout. This chapter describes tokenizer behavior on six datasets outside the ImageNet validation reference. ImageNet validation remains the in-distribution reference, and six shifted datasets vary sampling, visual style, object presentation, imaging domain, and document layout. The experiment measures changes in global code frequency, positional code diversity and reconstruction quality. These measurements describe how each quantity behaves under shift; they do not test whether a change in code usage causes a change in reconstruction quality.

Both tokenizers were trained on natural images. Chapter 4 evaluates them on the ImageNet validation set as the in-distribution reference. A shifted dataset can change their token distributions in two ways. First, it may activate codes that were absent on ImageNet, indicating that the ImageNet baseline did not expose the tokenizer’s operational vocabulary. Second, the active vocabulary may remain fixed while assignment probabilities change, concentrating many image patches onto a smaller effective set of codes. Raw active-code counts detect the first change but not the second.

Spatial allocation provides a separate view. Because each image is represented by a 16×16 grid, a code can be frequent globally yet appear in different parts of the grid under a new domain. Similarly, the range of codes observed at a fixed grid position may contract when a dataset has strong layout regularity. The chapter therefore separates three questions:

1. Does the global code-frequency distribution change?
2. Does code diversity change across grid positions?
3. How do the two checkpoints compare on reconstruction within each dataset?

Role	Dataset	Images
ID	ImageNet	50,000
OOD	ImageNet-V2	10,000
OOD	ObjectNet	50,273
OOD	ImageNet-Sketch	50,889
OOD	OrganAMNIST	57,187
OOD	BloodMNIST	17,092
OOD	RVL-CDIP	39,996

Table 8.1: Datasets encoded for the distribution-shift analysis. ImageNet is the in-distribution reference for every comparison. Export details are in Chapter 3.

8.1. Experimental Setup

8.1.1. Dataset-Shift Protocol

The fixed reference is the 50,000-image ImageNet validation set. It is compared separately with ImageNet-V2, ImageNet-Sketch, ObjectNet, OrganAMNIST, BloodMNIST, and RVL-CDIP, using the datasets and preprocessing described in Chapter 3. The comparison is always ImageNet versus one shifted dataset. Table 8.1 reports the number of encoded images. Both tokenizers produce a 16×16 grid, giving 256 token assignments per image.

The main conclusions about redistribution rely on ImageNet-Sketch, BloodMNIST, and RVL-CDIP, with ImageNet-V2 as a near-domain reference.

For each tokenizer and dataset, the experiment exports the global count of every code and a $K \times 16 \times 16$ tensor of code counts by grid position. The counts estimate code frequencies across the dataset and at each grid position.

8.1.2. Global Code-Usage Shift

Let p be the ImageNet code-frequency distribution defined in Section 3.3.3, and let q be the corresponding distribution on an OOD dataset. We define two ratios for this comparison, active-set retention and perplexity ratio, and use Jensen–Shannon divergence as a standard measure of distribution difference.

Active-set retention is defined as

$$\text{Ret}(p, q) = \frac{|\{k : p_k > 0\} \cap \{k : q_k > 0\}|}{|\{k : p_k > 0\}|}. \quad (8.1)$$

It measures the fraction of codes observed on the ID dataset that also occur on the OOD

dataset. Retention equals one if all ID-active codes remain active.

Perplexity is the exponential of distribution entropy, $P(p) = \exp(H(p))$. The **perplexity ratio** is defined as

$$R_P = \frac{P(q)}{P(p)} \quad (8.2)$$

and measures the change in effective vocabulary size. A value near one indicates similar spread; a value below one indicates concentration onto fewer effective codes.

The **Jensen–Shannon divergence** measures how much the probabilities of the same code identities differ between datasets. It is defined as:

$$\text{JSD}(p, q) = \frac{1}{2}D_{\text{KL}}(p\|m) + \frac{1}{2}D_{\text{KL}}(q\|m), \quad m = \frac{p+q}{2}. \quad (8.3)$$

Here m is the midpoint distribution. No sampling-null distribution was computed, so small nonzero estimates are not tests of a population shift. The vocabularies also differ between checkpoints; JSD magnitudes do not calibrate a cross-model robustness ranking. JSD is symmetric and remains finite when a code appears in only one dataset; direct KL divergence need not. It is zero for identical distributions and at most $\log 2$ with natural logarithms. Unlike a comparison between independently sorted rank-frequency curves, this quantity preserves code identity: probability assigned to code k on ImageNet is compared with probability assigned to the same code k on the shifted dataset. All entropy and divergence calculations use natural logarithms, so values are reported in nats. The comparison uses R_P to measure concentration relative to each tokenizer’s own ID baseline.

These global quantities pool millions of token assignments per dataset and are therefore less sparse than the per-code positional estimates below. Nevertheless, the datasets have unequal image counts, so small differences, especially for ImageNet-V2 and BloodMNIST, are not treated as formal hypothesis-test results.

8.1.3. Positional Measurements

For grid position (u, v) , let $p_{k|u,v}$ be the probability of observing code k at that position over all images in a dataset. Positional entropy is

$$H_{u,v} = - \sum_{k=1}^K p_{k|u,v} \log p_{k|u,v}. \quad (8.4)$$

We normalize entropy by $\log K$, its theoretical maximum for a codebook of size K . Both checkpoints have $K = 16,384$. The initial plots report the unadjusted average of $(H_{u,v}^{\text{shift}} - H_{u,v}^{\text{ImageNet}}) / \log K$ over all positions and separately over border and interior positions. Here the border is the one-cell outer ring of the 16×16 grid (60 positions); the remaining 196 positions form the interior. Negative values mean that fewer codes, or a more concentrated mix of codes, occur at that location under the shifted dataset.

These positional entropy estimates depend on sample size. Each position contributes one observation per image, so an empirical entropy computed from N images cannot exceed $\log(\min(K, N))$. ImageNet-V2 has 10,000 images, fewer than LlamaGen’s 16,384 codes. Its entropy difference consequently includes a sampling effect and cannot establish a loss of intrinsic diversity.

8.1.4. Reconstruction Measurements

PSNR, SSIM, LPIPS, and FID are computed between each dataset and its reconstructions under the protocol of Chapter 3. Paired metrics are reported consistently across rows, but their magnitudes remain sensitive to dataset content: a homogeneous medical dataset can yield high PSNR without being close to ImageNet. FID is used only to compare the two tokenizers on the same dataset. Its absolute value is not compared across datasets because each row has a different reference distribution. FID also inherits the natural-image bias of the Inception features on which it is based; this is particularly relevant for the medical and document datasets.

8.2. Experimental Results

8.2.1. The Active Vocabulary Is Stable but Its Distribution Is Not

Table 8.2 reports the global code-usage measurements. ImageNet-V2 produces almost no global shift: JSD is 0.002 for LlamaGen and 0.0004 for VQGAN, while both perplexity ratios are approximately one. The evaluated ObjectNet export also has low JSD values, 0.020 and 0.007. ImageNet-V2 contains natural images of the same classes collected from different sources.

The style and domain changes concentrate the token distributions. RVL-CDIP gives the largest JSD for both tokenizers, 0.251 for LlamaGen and 0.195 for VQGAN. BloodMNIST and ImageNet-Sketch follow. Their perplexity ratios fall to 0.359–0.377 for LlamaGen and

Tokenizer	Shifted dataset	Active retention	JSD	R_P
LlamaGen	ImageNet-V2	1.000	0.0020	0.995
LlamaGen	ObjectNet	1.000	0.0200	0.916
LlamaGen	ImageNet-Sketch	1.000	0.1531	0.377
LlamaGen	OrganAMNIST	1.000	0.1035	0.624
LlamaGen	BloodMNIST	0.984	0.2260	0.359
LlamaGen	RVL-CDIP	0.994	0.2510	0.230
VQGAN	ImageNet-V2	1.000	0.0004	1.001
VQGAN	ObjectNet	1.000	0.0068	0.962
VQGAN	ImageNet-Sketch	1.000	0.1277	0.452
VQGAN	OrganAMNIST	1.000	0.0424	0.816
VQGAN	BloodMNIST	0.994	0.1575	0.497
VQGAN	RVL-CDIP	0.999	0.1949	0.325

Table 8.2: Global code-usage shift relative to ImageNet. Active retention is the fraction of ImageNet-active codes also observed on the shifted dataset. Active retention is rounded to three decimals: LlamaGen uses 16,383 codes on ImageNet-Sketch and OrganAMNIST, so their displayed 1.000 is not exact. R_P is the shifted-to-ImageNet perplexity ratio.

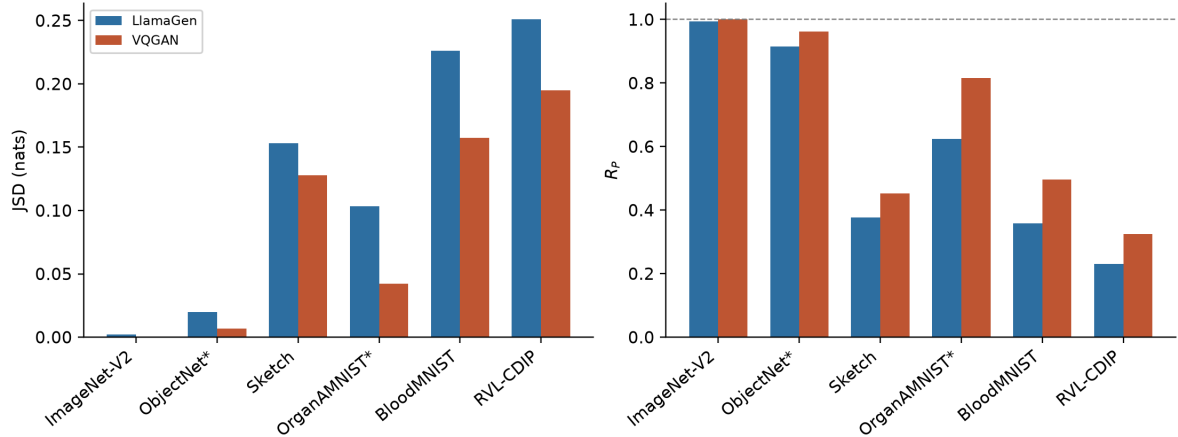


Figure 8.1: Global response to each shifted dataset. JSD measures movement of probability mass over fixed code identities; R_P measures concentration relative to each tokenizer’s ImageNet baseline. Asterisks mark the two exports detailed in Chapter 3.

0.452–0.497 for VQGAN. OrganAMNIST is intermediate. The ordering is broadly shared by the two models even though their ImageNet baselines are very different.

Active retention alone would miss this result. LlamaGen retains at least 98.4% of its 16,384 ImageNet-active codes on every dataset. VQGAN retains at least 99.4% of its 972 ImageNet-active codes, and exactly the same 972 codes remain active on ImageNet-V2, ImageNet-Sketch, ObjectNet, and OrganAMNIST. The shift therefore reallocates probability among an almost unchanged vocabulary; it does not activate ImageNet-inactive VQGAN codes at scale or collapse LlamaGen to a small raw set.

Figure 8.1 shows that ImageNet-V2 and the evaluated ObjectNet export remain near the ImageNet reference. The stylized, medical, and document datasets have lower perplexity

Tokenizer	Dataset	PSNR	SSIM	LPIPS	FID
LlamaGen	ImageNet	20.79	0.558	0.228	2.19
LlamaGen	ImageNet-V2	20.95	0.572	0.231	4.34
LlamaGen	ObjectNet	23.88	0.683	0.208	2.93
LlamaGen	ImageNet-Sketch	20.26	0.752	0.128	3.69
LlamaGen	OrganAMNIST	26.79	0.805	0.182	57.16
LlamaGen	BloodMNIST	27.10	0.802	0.157	19.60
LlamaGen	RVL-CDIP	20.49	0.764	0.094	5.08
VQGAN	ImageNet	19.65	0.489	0.286	5.06
VQGAN	ImageNet-V2	19.83	0.501	0.289	7.48
VQGAN	ObjectNet	22.79	0.611	0.268	5.80
VQGAN	ImageNet-Sketch	18.23	0.696	0.170	5.98
VQGAN	OrganAMNIST	26.58	0.732	0.239	76.12
VQGAN	BloodMNIST	26.38	0.766	0.201	29.02
VQGAN	RVL-CDIP	18.23	0.699	0.128	8.02

Table 8.3: Reconstruction quality on ImageNet and the shifted datasets. PSNR and SSIM are higher-is-better; LPIPS and FID are lower-is-better. FID comparisons are made between tokenizers within a dataset, not across dataset rows. PSNR is in dB.

ratios, despite little change in their raw active sets.

8.2.2. Reconstruction under Distribution Shift

Table 8.3 reports reconstruction quality on the six shifted datasets. LlamaGen is better in every model-paired comparison: its PSNR and SSIM are higher, while its LPIPS and FID are lower. The advantage is not confined to the ImageNet baseline established in Chapter 4.

The paired reconstruction metrics vary across the shifted datasets. ImageNet-V2 has similar scores to ImageNet for both tokenizers. ImageNet-Sketch has lower PSNR but higher SSIM and lower LPIPS, so the metrics do not support a single ordering of those datasets. Medical images have higher PSNR and SSIM and lower LPIPS than ImageNet, while document images have higher SSIM and lower LPIPS but lower PSNR. These differences also reflect image content and complexity. The consistent comparison is between tokenizers on each dataset: all four metrics favor LlamaGen.

The different code-frequency profiles do not correspond to a uniform pattern in the reported reconstruction metrics. Scores across datasets are not a controlled measure of robustness, however, because image content and complexity also change. FID is reported for the model comparison within each dataset; its changing reference distribution and natural-image features limit interpretation across domains.

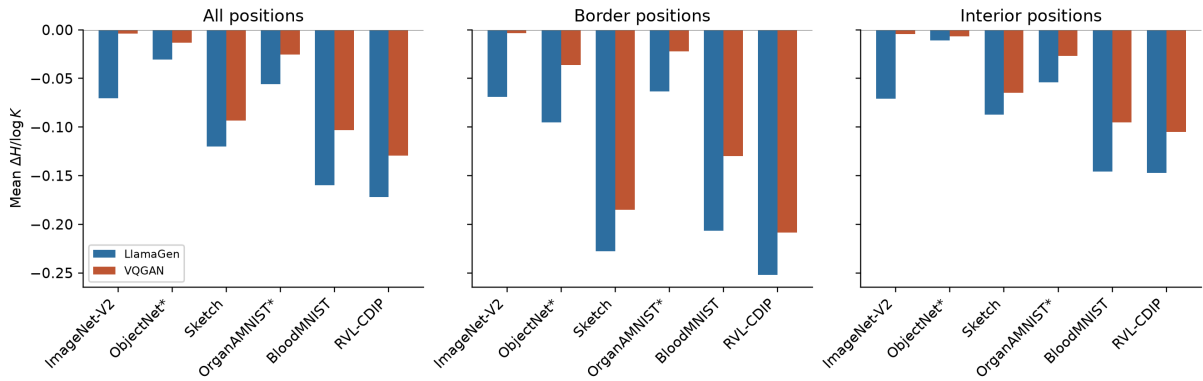


Figure 8.2: Mean positional entropy change divided by $\log K$, for all positions, border positions, and interior positions. The common scale expresses changes relative to the theoretical entropy range. These are full-reference, unequal-size estimates; Table 8.4 checks the three smaller datasets with equal-size ImageNet references. Asterisks mark the two exports detailed in Chapter 3.

8.2.3. Positional Entropy Estimates under Shift

Figure 8.2 reports normalized changes in positional entropy on a common scale. Negative values indicate a more concentrated distribution at a typical grid position. Comparing border and interior averages tests whether that concentration is uniform across the grid. For ImageNet-Sketch, LlamaGen’s mean change is -2.21 nats on the border versus -0.85 nats in the interior; for RVL-CDIP, it is -2.45 versus -1.43 nats. The evaluated ObjectNet export also shows a larger border than interior change (-0.92 versus -0.11 nats). These averages describe spatial variation, including effects of margins, backgrounds, and preprocessing. Both panels divide by $\log 16,384$ for a common scale, although VQGAN’s empirical ImageNet vocabulary of 972 codes has a smaller attainable range. Normalized bar heights therefore are not directly comparable as fractions of each checkpoint’s empirical range. Table C.14 gives the full all/border/interior values in nats, without normalization.

The ImageNet-V2 result illustrates why these maps require caution. Its global distribution is almost unchanged, but LlamaGen’s estimated mean positional entropy is 0.68 nats lower (0.070 after division by $\log K$). ImageNet-V2 has only 10,000 images against 50,000 in the reference set. To separate this sampling effect from the dataset contrast, we drew 20 ImageNet subsets without replacement, each the same size as the shifted dataset, and recomputed positional entropy from the archived per-image clean token grids. These grids reproduce the original ImageNet position counts exactly. Table 8.4 reports the full-reference and sample-matched estimates in nats.

Tokenizer	Dataset	Full reference	Matched reference
LlamaGen	ImageNet-V2	-0.6825	-0.0036
LlamaGen	BloodMNIST	-1.5519	-1.1849
LlamaGen	RVL-CDIP	-1.6679	-1.6181
VQGAN	ImageNet-V2	-0.0390	+0.0006
VQGAN	BloodMNIST	-1.0034	-0.9846
VQGAN	RVL-CDIP	-1.2541	-1.2516

Table 8.4: Mean positional entropy difference from ImageNet in nats, before and after matching the ImageNet reference count to each shifted dataset. Matched values average 20 subsets. The matching changes the reference sample size, not the tokenizers or shifted-dataset observations.

The apparent ImageNet-V2 entropy decrease nearly vanishes for both models after matching. The 2.5th and 97.5th percentiles across the 20 reference draws are -0.0044 and -0.0026 nats for LlamaGen, and -0.0002 and $+0.0018$ for VQGAN. These percentiles describe reference-subset variation, not confidence intervals for a population-level shift. BloodMNIST and RVL-CDIP retain substantial negative differences after matching, so their patterns cannot be explained by this sample-count mismatch alone. ImageNet-Sketch, ObjectNet, and OrganAMNIST each have more than 50,000 images. With only aggregate OOD position counts available for them, their plotted differences remain descriptive rather than sample-matched.

8.3. Discussion

The distribution-shift experiment separates vocabulary availability from vocabulary use. VQGAN’s approximately 972 active ImageNet codes remain almost exactly the active set on all six shifted datasets. LlamaGen likewise retains nearly its full vocabulary. Neither tokenizer responds to a new domain by substantially expanding or shrinking its raw active set. The measurable change is redistribution: frequent codes become more dominant and perplexity falls. Raw active-code count detects changes in vocabulary membership, but it misses this redistribution within the active set.

ImageNet-V2 remains close to the ImageNet token distribution. ImageNet-Sketch, BloodMNIST, and RVL-CDIP use a more concentrated token mixture, with RVL-CDIP producing the largest JSD. The two tokenizers show the same qualitative JSD ranking despite their different empirical vocabulary sizes.

Changing the input domain substantially changes token frequencies for sketches, medical images, and documents, while the reported reconstruction scores show no uniform ordering across these different datasets. Each tokenizer produces reconstructions on these datasets, but differences in image content prevent a causal reading of cross-dataset score changes.

The global results are the main finding of this chapter. An almost unchanged active vocabulary can hide substantial redistribution among its codes. LlamaGen retains its reconstruction advantage across the tested datasets, but its larger vocabulary does not prevent this redistribution.

9 | Conclusion

This thesis compared VQGAN and LlamaGen VQ-16 to determine how two pretrained image tokenizers with the same 16×16 output grid differ beyond reconstruction quality. The study measured reconstruction, empirical codebook usage, token stability under global noise, encoder and decoder locality, and the response to distribution shift. Each analysis isolated the tokenizer from the autoregressive model that would consume its tokens.

For RQ1, the reconstruction analysis established the clean baseline under each exporter’s recorded preprocessing. On ImageNet validation, LlamaGen achieved 20.79 rather than 19.65 dB PSNR, 0.558 rather than 0.489 SSIM, 0.228 rather than 0.286 LPIPS, and 2.19 rather than 5.06 FID. The codebook measurements revealed a larger difference between the checkpoints. LlamaGen used all 16,384 nominal entries at least once, while VQGAN used about 972. The nominal vocabulary size therefore did not describe the vocabulary used by each checkpoint.

For RQ2, global image perturbations showed that neither tokenizer preserved exact token identity under the tested Gaussian noise. At $\sigma = 0.1$ in the internal $[-1, 1]$ image range (equivalent to a standard deviation of 0.05 in $[0, 1]$ before clipping), 79.5% of LlamaGen positions and 75.1% of VQGAN positions differed from their clean assignments. LlamaGen had higher PSNR to the original clean image at every tested noise level. A separate 2,000-image comparison of saved clean and noisy reconstructions found lower decoded RMSE for LlamaGen at all three noise levels, despite its slightly higher identity-flip fraction. The full-dataset PSNR-to-original measure and this sampled direct-change measure answer different questions. Noise also concentrated code usage in both models without greatly changing the raw number of active codes.

For RQ3, local image perturbations showed that the encoder response was not confined to the perturbed patch. At low patch noise, the far-field token flip rate was 17.4% for VQGAN and 17.8% for LlamaGen. Masking one image patch changed 56.3% of VQGAN positions and 49.5% of LlamaGen positions outside the corresponding token region.

For RQ4, the decoder analysis found that directly editing a local token patch produced

the largest pixel changes near the edited region, although measurable changes remained outside it. The same pattern held on the slightly out-of-distribution ImageNet-V2 dataset, where the largest change in patch LPIPS relative to ImageNet validation was 0.0068. Locality is therefore preserved approximately rather than exactly when the decoder maps token edits back to pixels. A saved-image distance profile for the 25% edit shows that outside response declines with distance from the edited boundary but remains measurable 80–96 pixels away on the sampled images.

For RQ5, both VQGAN and LlamaGen retained almost all codes that were active on ImageNet. They did not switch to a substantially different set of active tokens. Instead, they changed how often they used those tokens. ImageNet-V2 remained close to the ImageNet token distribution, while ImageNet-Sketch, BloodMNIST, and RVL-CDIP produced larger changes. RVL-CDIP produced the largest Jensen–Shannon divergence for both tokenizers. The reconstruction scores do not show a uniform ordering across these datasets, but their different image content prevents that comparison from isolating the effect of distribution shift on reconstruction quality.

Reconstruction, token identity, and code-frequency measurements describe different properties of these checkpoints. High flip rates occurred in both systems, but do not calibrate a robustness ranking across their different codebooks. The distribution-shift results show that an almost unchanged active set can conceal substantial frequency changes. A tokenizer therefore determines more than how faithfully an image is rebuilt; it also determines the sequence distribution presented to the autoregressive model.

9.1. Limitations and Future Work

This study compared one pretrained VQGAN checkpoint with one pretrained LlamaGen VQ-16 checkpoint. They share a VQGAN-style encoder–quantizer–decoder backbone, but differ in checkpoint, training recipe, codebook dimensionality, normalization, and implementation details. The observed differences describe the two complete systems. They do not identify which design choice caused each result. A direct mechanistic study should modify one tokenizer property at a time, retrain the affected models, and repeat the same measurements. This would test the effects of codebook geometry and training choices while controlling the remaining components.

All experiments in this thesis use 256×256 inputs. Their exporters used different resize filters, as documented in the reproducibility appendix. Resizing and cropping can alter datasets whose native resolution, aspect ratio, or layout differs from ImageNet, especially document and medical images. Future comparisons should evaluate the existing check-

points at other supported resolutions and test how the distribution-shift patterns depend on the chosen input size.

ObjectNet retained its identification border, and the evaluated OrganAMNIST export omitted 1,643 training images (Section 3.2.2); a replication should correct both inputs.

The analyses of global and local perturbations used the same seeds and intervention levels for both models, but they did not save a shared record of the exact noise arrays and patch locations. Identical seed values do not imply identical perturbations across separate implementations and preprocessing pipelines; the realizations and patch locations were not matched. A replication should export each intervention once and apply it to both tokenizers. The global-noise sweep starts at $\sigma = 0.1$ in the internal image range; it does not characterize weaker perturbations. Clipping changes the sampled noise distribution, particularly at the strongest level. Decoder edit modes also differ in induced distance between codebooks, and VQGAN random edits can select codes absent from clean ImageNet assignments. These interventions do not identify intrinsic cross-model sensitivity or individual causal mechanisms (Sections 5.1.1 and 7.3).

The clean ImageNet PSNR comparison has a paired image-level bootstrap interval and win rate; the other reconstruction metrics do not have analogous paired estimates. The decoder distance-to-boundary profile uses 500 saved eight-bit image pairs per checkpoint at one patch size; it describes spatial decay, not a fixed-width halo or an exact receptive-field radius. The archived per-image ImageNet grids allowed a sample-matched positional-entropy check for ImageNet-V2, BloodMNIST, and RVL-CDIP. Per-image grids are not available for every shifted dataset, so the remaining comparisons could not receive the same check. No sampling null was computed for JSD, so small divergences are descriptive rather than significance tests. Cross-dataset reconstruction scores also depend on image content; they do not establish preservation or loss of quality caused by a distribution shift.

Finally, the study did not train or evaluate an autoregressive model on the measured token streams. It therefore cannot show how token instability, concentrated code usage, or distribution-dependent token frequencies affect generation. Future work should train or evaluate autoregressive priors on these streams and measure generation quality and robustness directly. Further spatial analysis should retain per-image grids for every dataset so all positional-entropy comparisons can use matched counts.

The shared 16×16 grid and 16,384-entry nominal codebook concealed very different empirical vocabularies. Both checkpoints changed token identities under the tested noise, responded beyond local intervention regions, and redistributed code use across image domains. Reconstruction scores alone did not reveal these behaviors.

A | Implementation and Code Organization

A.1. Repository Structure

The experimental implementation is divided across three repositories. The `taming-transformers` repository contains the VQGAN model and its experiment scripts [7], while `LlamaGen` contains the corresponding tokenizer implementation [11]. The `vq-explore` repository contains the comparative analysis notebooks and the decoder-locality analysis utility. This separation keeps model-specific inference code close to each model while placing cross-model analysis in one location.

The repositories are available at <https://github.com/farzinnasiri/taming-transformers>, <https://github.com/farzinnasiri/LlamaGen>, and <https://github.com/farzinnasiri/vq-explore>. The thesis source and compact results are available at <https://github.com/farzinnasiri/Thesis---VQ>.

A.2. Experiment Scripts

Equivalent scripts were implemented for the two tokenizers. The reconstruction scripts encode and decode image-folder datasets and export per-image fidelity metrics and reconstructed images. The code-usage scripts accumulate global and position-dependent token counts. The robustness exporters implement the three interventions used in this thesis: global Gaussian noise, local image-space patch perturbations, and local token-space replacement.

For each robustness run, images are processed in batches on the GPU. Image writing and metadata export are delegated to worker processes so that disk operations do not block inference. Each sample is stored under its dataset class and image identifier. Sharded JSONL metadata records the perturbation parameters, patch coordinates, and clean and perturbed token grids. This format allows later analyses to recompute metrics without

rerunning either tokenizer.

The token-space experiments additionally require mappings from each token to a closest, farthest, and approximately orthogonal alternative. LlamaGen mappings are computed over its full, L2-normalized codebook. VQGAN mappings are computed over the empirically active subset used by the experiment, as described in Chapter 7.

A.3. Analysis Notebooks and Outputs

One notebook consolidates the analysis for each experimental chapter:

- `ch04_rq1_reconstruction_codebook_usage.ipynb`;
- `ch05_rq2_global_robustness.ipynb`;
- `ch06_rq3_local_image_perturbations.ipynb`;
- `ch07_rq4_token_space_perturbations.ipynb`;
- `ch08_rq5_distribution_shift.ipynb`.

The notebooks read the exported images and metadata, compute chapter-level summaries, and save the tables and figures cited in the thesis. Expensive per-image calculations are cached as CSV, Parquet, or NPZ files where appropriate. CPU-bound image metrics are parallelized across workers, while LPIPS is evaluated in GPU batches.

On the analysis host, chapter outputs follow a common layout:

- `chapterN_outputs/data`: summary tables;
- `chapterN_outputs/figures`: plots;
- `chapterN_outputs/qualitative_samples`: source images used for visual comparisons.

Compact source artifacts are copied into `Assets/rqN/data` and `Assets/rqN/figures` in the thesis repository, where `rqN` denotes the research-question number. Figures included by LaTeX are copied into `Images`. Raw reconstructions and complete run exports remain outside the thesis repository.

The linked repositories contain source and compact results. The complete versions of the code and notebooks, saved run data, and reproduction instructions will be provided separately to the supervisor.

A.4. Experimental Configuration

Both tokenizers process RGB images at 256×256 resolution using a shortest-side resize followed by a center crop. Both expose a nominal 16,384-entry codebook and produce a 16×16 token grid. The VQGAN runs `load /checkpoints/vqgan_imagenet_f16_16384/model.yaml` and `last.ckpt`; the LlamaGen runs `load /checkpoints/vq_ds16_c2i.pt` with the VQ-16 architecture, embedding dimension 8, and codebook size 16,384. Robustness exporters set the Python, NumPy, and PyTorch random seeds to 0. Code-usage exports are deterministic with respect to input order: class directories and file names are sorted, data loading uses `shuffle=False`, and no token sampling is performed.

For each code-usage run, `global_counts.npy` stores one count per codebook index and `position_counts.npy` stores counts with shape $K \times 16 \times 16$. The exporter also writes `usage.csv`, the 500 most frequent codes, and `summary.json`. When enabled, `indices.npz` stores one token grid and source path per image. The Chapter 8 artifact inventory verifies image counts, token-grid shape, total token counts, and active-code counts for every model–dataset pair. These checks are archived in `Assets/rq5/data/ch08_artifact_inventory.csv`.

B | Computational Environment

B.1. Containerized Execution

The two tokenizer implementations have incompatible software requirements and were therefore executed in separate Docker images on a CUDA-enabled server. The ImageNet datasets and pretrained checkpoints were mounted into the containers, while experiment directories were mounted into persistent storage. Long-running full-dataset exports were launched in background terminal sessions. Random seeds, perturbation modes, patch sizes, and output locations were fixed by script arguments or environment variables and recorded in run directory names and metadata.

B.2. Tokenizer Environments

The VQGAN environment is based on Ubuntu 20.04 with CUDA 11.7, Python 3.8, PyTorch 1.13.1, and torchvision 0.14.1. It retains the older dependencies required by Taming Transformers, including PyTorch Lightning 1.0.8, OmegaConf 2.0.0, and scikit-image 0.18.3.

The LlamaGen environment uses Ubuntu 20.04 with CUDA 12.2, PyTorch 2.1.2, and torchvision 0.16.2. Its analysis dependencies include NumPy, Pillow, scikit-image, pandas, Matplotlib, and LPIPS. Both environments expose the same dataset and checkpoint mounts, allowing the model-specific exporters to produce compatible image and metadata layouts.

The model exporters do not use byte-identical resizing operators. LlamaGen repeatedly downsamples large images with Pillow BOX filtering, resizes the shortest side with bicubic interpolation, and then applies an integer center crop. The VQGAN exporter resizes the shortest side with Lanczos interpolation before center cropping. Both produce RGB 256×256 tensors, but this filtering difference is a preprocessing confound in fine-grained cross-model comparisons.

B.3. Evaluation and Analysis Environment

Distribution-level reconstruction metrics are computed in a separate evaluator image containing TensorFlow CPU 2.9, PyTorch, torchvision, and `torch-fidelity`. Reconstructions are exported as uint8 NPZ arrays and compared with reference statistics generated from the same ImageNet validation preprocessing. Using a separate evaluator avoids changing either tokenizer environment solely to satisfy metric dependencies.

The chapter notebooks run in the Jupyter environment associated with `vq-explore`. They use pandas and NumPy for aggregation, Matplotlib for figures, scikit-image for pixel metrics, and batched LPIPS evaluation where required. Full ImageNet experiments use 50,000 validation images. DataLoader workers, asynchronous image writers, process pools, and GPU metric batches are used to keep model inference and analysis practical without changing the experimental definitions.

C | Supporting Experimental Results

The tables below retain the per-condition summaries behind the figures and selected comparisons in Chapters 5–8. A dash marks a measurement missing from the archived summary, not an estimated value. In particular, the LlamaGen 10% closest-edit patch PSNR mean and standard deviation are absent from the saved fidelity summary. Their cause cannot be recovered from that summary, so no replacement value is inferred. Pixel RMSE uses RGB values in $[0, 1]$. Standard deviations describe image-to-image variation and are not confidence intervals. The model runs used the same nominal settings but did not save shared random interventions; cross-model values are descriptive comparisons.

C.1. Global image noise

Table C.1 records fidelity to the noisy input, which has a different reference from the complete summary that follows.

Tokenizer	$\sigma = 0.1$	$\sigma = 0.2$	$\sigma = 0.5$
LlamaGen	19.26	16.76	11.87
VQGAN	18.36	16.11	11.31

Table C.1: Mean PSNR($x_\sigma, \hat{x}_\sigma$) in dB, from the archived global-noise PSNR summary.

Table C.2: Complete ImageNet global-noise summary. PSNR values are dataset means in dB; flip is the fraction of changed token assignments, and top-100 mass is the assignment probability held by the 100 most frequent codes.

Tokenizer	σ	PSNR($x, \hat{x}_\sigma$)	Flip	Perplexity	Top-100 mass	Active codes
LlamaGen	0.00	20.79	0.000	16179	0.012	16,384
LlamaGen	0.10	20.32	0.795	12829	0.040	16,381
LlamaGen	0.20	19.11	0.910	10388	0.063	16,369
LlamaGen	0.50	16.22	0.979	6434	0.115	16,203

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Tokenizer	σ	PSNR($x, \hat{x}_\sigma$)	Flip	Perplexity	Top-100 mass	Active codes
VQGAN	0.00	19.65	0.000	910	0.165	972
VQGAN	0.10	19.23	0.751	859	0.202	973
VQGAN	0.20	18.11	0.890	780	0.249	971
VQGAN	0.50	14.90	0.975	560	0.388	972

C.2. Encoder response by distance from the patch

Table C.3: Token-flip probability and pooled token count N_d by Manhattan distance from the patch. Counts are identical in the six saved model/noise conditions. Every value is taken from the saved Chapter 6 distance profile. Images with no tokens at a distance contribute nothing to that row.

Distance	N_d	VQ 0.1	VQ 0.25	VQ 0.5	Llama 0.1	Llama 0.25	Llama 0.5
0	3200000	0.788	0.952	0.988	0.822	0.960	0.991
1	1422912	0.597	0.827	0.917	0.613	0.832	0.922
2	1404408	0.420	0.634	0.741	0.426	0.614	0.710
3	1345483	0.295	0.448	0.536	0.285	0.413	0.494
4	1247461	0.226	0.339	0.411	0.221	0.317	0.385
5	1117010	0.196	0.294	0.360	0.196	0.281	0.346
6	951939	0.184	0.277	0.343	0.187	0.269	0.335
7	758797	0.178	0.270	0.336	0.182	0.263	0.331
8	539705	0.171	0.260	0.328	0.176	0.257	0.326
9	295818	0.164	0.255	0.324	0.169	0.250	0.321
10	206889	0.162	0.252	0.323	0.167	0.246	0.320
11	137713	0.160	0.248	0.318	0.166	0.248	0.321
12	85927	0.161	0.249	0.319	0.164	0.246	0.320
13	49090	0.157	0.245	0.316	0.165	0.245	0.320
14	24505	0.158	0.243	0.312	0.165	0.246	0.323
15	9855	0.155	0.242	0.313	0.175	0.256	0.330
16	2488	0.154	0.233	0.291	0.160	0.242	0.316

C.3. Token edits on ImageNet validation

Table C.4: ImageNet validation token-edit fidelity over the patch. Patch fraction is the target fraction of the 16×16 token grid; PSNR is in dB. All metrics compare the edited reconstruction with the clean reconstruction.

Tokenizer	Patch	Edit	PSNR	SSIM	LPIPS
LlamaGen	10%	Closest	–	0.629	0.109
LlamaGen	10%	Orthogonal	11.95	0.139	0.519
LlamaGen	10%	Random	11.97	0.150	0.511
LlamaGen	10%	Farthest	8.42	0.035	0.630

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Tokenizer	Patch	Edit	PSNR	SSIM	LPIPS
LlamaGen	25%	Closest	20.36	0.622	0.134
LlamaGen	25%	Orthogonal	11.45	0.129	0.599
LlamaGen	25%	Random	11.48	0.138	0.592
LlamaGen	25%	Farthest	7.74	0.027	0.702
LlamaGen	50%	Closest	20.44	0.629	0.144
LlamaGen	50%	Orthogonal	11.24	0.129	0.638
LlamaGen	50%	Random	11.27	0.138	0.633
LlamaGen	50%	Farthest	7.28	0.024	0.742
LlamaGen	75%	Closest	20.53	0.631	0.154
LlamaGen	75%	Orthogonal	10.85	0.120	0.675
LlamaGen	75%	Random	10.88	0.130	0.671
LlamaGen	75%	Farthest	6.61	0.019	0.782
VQGAN	10%	Closest	20.44	0.598	0.139
VQGAN	10%	Orthogonal	11.14	0.110	0.537
VQGAN	10%	Random	13.41	0.210	0.598
VQGAN	10%	Farthest	3.96	0.050	0.730
VQGAN	25%	Closest	20.00	0.579	0.182
VQGAN	25%	Orthogonal	10.66	0.098	0.614
VQGAN	25%	Random	12.83	0.232	0.749
VQGAN	25%	Farthest	3.74	0.050	0.790
VQGAN	50%	Closest	19.98	0.579	0.206
VQGAN	50%	Orthogonal	10.41	0.096	0.655
VQGAN	50%	Random	12.53	0.258	0.823
VQGAN	50%	Farthest	3.67	0.049	0.828
VQGAN	75%	Closest	19.92	0.573	0.230
VQGAN	75%	Orthogonal	10.01	0.088	0.697
VQGAN	75%	Random	12.08	0.269	0.879
VQGAN	75%	Farthest	3.64	0.046	0.861

Table C.5: ImageNet validation token-edit fidelity over the full image. Patch fraction is the target fraction of the 16×16 token grid; PSNR is in dB. All metrics compare the edited reconstruction with the clean reconstruction.

Tokenizer	Patch	Edit	PSNR	SSIM	LPIPS
LlamaGen	10%	Closest	29.34	0.944	0.025
LlamaGen	10%	Orthogonal	20.83	0.838	0.116
LlamaGen	10%	Random	20.86	0.841	0.115

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Tokenizer	Patch	Edit	PSNR	SSIM	LPIPS
LlamaGen	10%	Farthest	17.58	0.807	0.151
LlamaGen	25%	Closest	25.53	0.874	0.057
LlamaGen	25%	Orthogonal	16.74	0.656	0.261
LlamaGen	25%	Random	16.77	0.660	0.258
LlamaGen	25%	Farthest	13.22	0.600	0.324
LlamaGen	50%	Closest	23.13	0.784	0.097
LlamaGen	50%	Orthogonal	13.99	0.433	0.442
LlamaGen	50%	Random	14.03	0.441	0.438
LlamaGen	50%	Farthest	10.18	0.352	0.532
LlamaGen	75%	Closest	21.42	0.687	0.138
LlamaGen	75%	Orthogonal	11.76	0.219	0.617
LlamaGen	75%	Random	11.79	0.228	0.613
LlamaGen	75%	Farthest	7.60	0.122	0.723
VQGAN	10%	Closest	28.30	0.929	0.034
VQGAN	10%	Orthogonal	19.65	0.799	0.141
VQGAN	10%	Random	21.29	0.806	0.136
VQGAN	10%	Farthest	12.95	0.726	0.221
VQGAN	25%	Closest	24.60	0.842	0.080
VQGAN	25%	Orthogonal	15.77	0.604	0.295
VQGAN	25%	Random	17.38	0.609	0.321
VQGAN	25%	Farthest	9.03	0.474	0.440
VQGAN	50%	Closest	22.27	0.734	0.139
VQGAN	50%	Orthogonal	13.09	0.378	0.480
VQGAN	50%	Random	14.82	0.425	0.557
VQGAN	50%	Farthest	6.42	0.243	0.672
VQGAN	75%	Closest	20.63	0.625	0.206
VQGAN	75%	Orthogonal	10.91	0.175	0.651
VQGAN	75%	Random	12.82	0.308	0.788
VQGAN	75%	Farthest	4.54	0.091	0.839

Table C.6: ImageNet validation decoder locality after token edits. Inside and outside are mean pixel RMSE relative to the clean reconstruction; leakage is the median per-image outside-to-inside ratio. The 99th percentile is the mean across images of each image’s outside-patch 99th percentile.

Tokenizer	Patch	Edit	Inside	Outside	Leakage	Outside P99
LlamaGen	10%	Closest	0.067	0.008	0.122	0.080
LlamaGen	10%	Orthogonal	0.210	0.023	0.111	0.198
LlamaGen	10%	Random	0.208	0.023	0.111	0.196
LlamaGen	10%	Farthest	0.329	0.030	0.091	0.256
LlamaGen	25%	Closest	0.067	0.013	0.193	0.105
LlamaGen	25%	Orthogonal	0.222	0.039	0.176	0.251
LlamaGen	25%	Random	0.220	0.038	0.175	0.250
LlamaGen	25%	Farthest	0.356	0.050	0.141	0.328
LlamaGen	50%	Closest	0.066	0.019	0.292	0.131
LlamaGen	50%	Orthogonal	0.229	0.061	0.267	0.303
LlamaGen	50%	Random	0.226	0.060	0.267	0.302
LlamaGen	50%	Farthest	0.376	0.078	0.210	0.403
LlamaGen	75%	Closest	0.065	0.027	0.418	0.157
LlamaGen	75%	Orthogonal	0.242	0.092	0.385	0.352
LlamaGen	75%	Random	0.240	0.091	0.383	0.352
LlamaGen	75%	Farthest	0.410	0.120	0.299	0.477
VQGAN	10%	Closest	0.070	0.011	0.166	0.110
VQGAN	10%	Orthogonal	0.227	0.029	0.126	0.258
VQGAN	10%	Random	0.187	0.030	0.159	0.235
VQGAN	10%	Farthest	0.592	0.053	0.088	0.549
VQGAN	25%	Closest	0.072	0.019	0.263	0.148
VQGAN	25%	Orthogonal	0.239	0.046	0.187	0.317
VQGAN	25%	Random	0.200	0.050	0.249	0.302
VQGAN	25%	Farthest	0.607	0.090	0.147	0.673
VQGAN	50%	Closest	0.072	0.028	0.390	0.183
VQGAN	50%	Orthogonal	0.247	0.067	0.270	0.373
VQGAN	50%	Random	0.207	0.075	0.360	0.352
VQGAN	50%	Farthest	0.611	0.140	0.231	0.746
VQGAN	75%	Closest	0.074	0.039	0.531	0.216
VQGAN	75%	Orthogonal	0.261	0.097	0.375	0.425
VQGAN	75%	Random	0.220	0.104	0.485	0.382
VQGAN	75%	Farthest	0.613	0.213	0.352	0.780

Table C.7: ImageNet validation variation across images for each token-edit condition. Columns are the saved per-condition standard deviations: patch PSNR in dB, patch LPIPS, and inside and outside pixel RMSE.

Tokenizer	Patch	Edit	PSNR SD	LPIPS SD	Inside SD	Outside SD
LlamaGen	10%	Closest	–	0.041	0.026	0.003
LlamaGen	10%	Orthogonal	2.23	0.081	0.056	0.007
LlamaGen	10%	Random	2.32	0.081	0.058	0.007
LlamaGen	10%	Farthest	2.16	0.100	0.092	0.008
LlamaGen	25%	Closest	3.02	0.040	0.023	0.005
LlamaGen	25%	Orthogonal	1.76	0.067	0.049	0.010
LlamaGen	25%	Random	1.82	0.068	0.050	0.010
LlamaGen	25%	Farthest	1.83	0.086	0.086	0.013
LlamaGen	50%	Closest	2.75	0.040	0.021	0.007
LlamaGen	50%	Orthogonal	1.57	0.062	0.046	0.015
LlamaGen	50%	Random	1.62	0.061	0.046	0.015
LlamaGen	50%	Farthest	1.66	0.080	0.083	0.020
LlamaGen	75%	Closest	2.59	0.041	0.020	0.011
LlamaGen	75%	Orthogonal	1.51	0.061	0.049	0.025
LlamaGen	75%	Random	1.55	0.061	0.049	0.025
LlamaGen	75%	Farthest	1.58	0.079	0.087	0.035
VQGAN	10%	Closest	3.88	0.052	0.028	0.005
VQGAN	10%	Orthogonal	2.05	0.088	0.056	0.010
VQGAN	10%	Random	2.62	0.103	0.058	0.011
VQGAN	10%	Farthest	1.34	0.089	0.104	0.014
VQGAN	25%	Closest	3.16	0.053	0.025	0.008
VQGAN	25%	Orthogonal	1.60	0.075	0.047	0.014
VQGAN	25%	Random	2.17	0.082	0.054	0.017
VQGAN	25%	Farthest	1.13	0.078	0.089	0.022
VQGAN	50%	Closest	2.85	0.054	0.024	0.012
VQGAN	50%	Orthogonal	1.39	0.069	0.043	0.020
VQGAN	50%	Random	1.98	0.075	0.052	0.023
VQGAN	50%	Farthest	1.08	0.076	0.084	0.029
VQGAN	75%	Closest	2.65	0.056	0.023	0.017
VQGAN	75%	Orthogonal	1.26	0.069	0.043	0.029
VQGAN	75%	Random	1.93	0.075	0.056	0.030
VQGAN	75%	Farthest	1.09	0.076	0.086	0.049

C.4. Token edits on ImageNet-V2

Table C.8: ImageNet-V2 token-edit fidelity over the patch. Patch fraction is the target fraction of the 16×16 token grid; PSNR is in dB. All metrics compare the edited reconstruction with the clean reconstruction.

Tokenizer	Patch	Edit	PSNR	SSIM	LPIPS
LlamaGen	10%	Closest	20.72	0.634	0.110
LlamaGen	10%	Orthogonal	11.92	0.139	0.513
LlamaGen	10%	Random	11.93	0.149	0.505
LlamaGen	10%	Farthest	8.29	0.031	0.625
LlamaGen	25%	Closest	20.44	0.628	0.136
LlamaGen	25%	Orthogonal	11.43	0.128	0.592
LlamaGen	25%	Random	11.45	0.139	0.586
LlamaGen	25%	Farthest	7.59	0.024	0.698
LlamaGen	50%	Closest	20.54	0.636	0.146
LlamaGen	50%	Orthogonal	11.22	0.129	0.632
LlamaGen	50%	Random	11.22	0.138	0.627
LlamaGen	50%	Farthest	7.11	0.021	0.740
LlamaGen	75%	Closest	20.65	0.638	0.155
LlamaGen	75%	Orthogonal	10.82	0.121	0.670
LlamaGen	75%	Random	10.85	0.130	0.666
LlamaGen	75%	Farthest	6.45	0.015	0.780
VQGAN	10%	Closest	20.52	0.605	0.140
VQGAN	10%	Orthogonal	11.15	0.111	0.531
VQGAN	10%	Random	13.38	0.210	0.596
VQGAN	10%	Farthest	3.94	0.048	0.726
VQGAN	25%	Closest	20.10	0.587	0.183
VQGAN	25%	Orthogonal	10.68	0.099	0.608
VQGAN	25%	Random	12.82	0.233	0.749
VQGAN	25%	Farthest	3.71	0.047	0.786
VQGAN	50%	Closest	20.10	0.588	0.207
VQGAN	50%	Orthogonal	10.42	0.097	0.649
VQGAN	50%	Random	12.51	0.259	0.822
VQGAN	50%	Farthest	3.64	0.047	0.822
VQGAN	75%	Closest	20.05	0.581	0.232
VQGAN	75%	Orthogonal	10.01	0.089	0.691
VQGAN	75%	Random	12.04	0.270	0.876
VQGAN	75%	Farthest	3.60	0.044	0.854

Table C.9: ImageNet-V2 token-edit fidelity over the full image. Patch fraction is the target fraction of the 16×16 token grid; PSNR is in dB. All metrics compare the edited reconstruction with the clean reconstruction.

Tokenizer	Patch	Edit	PSNR	SSIM	LPIPS
LlamaGen	10%	Closest	29.41	0.945	0.025
LlamaGen	10%	Orthogonal	20.85	0.841	0.117
LlamaGen	10%	Random	20.87	0.843	0.115
LlamaGen	10%	Farthest	17.50	0.809	0.151
LlamaGen	25%	Closest	25.63	0.877	0.058
LlamaGen	25%	Orthogonal	16.74	0.660	0.260
LlamaGen	25%	Random	16.77	0.664	0.257
LlamaGen	25%	Farthest	13.10	0.603	0.324
LlamaGen	50%	Closest	23.25	0.789	0.098
LlamaGen	50%	Orthogonal	13.98	0.437	0.440
LlamaGen	50%	Random	13.99	0.444	0.436
LlamaGen	50%	Farthest	10.03	0.353	0.532
LlamaGen	75%	Closest	21.55	0.694	0.139
LlamaGen	75%	Orthogonal	11.74	0.222	0.613
LlamaGen	75%	Random	11.77	0.231	0.608
LlamaGen	75%	Farthest	7.44	0.120	0.721
VQGAN	10%	Closest	28.46	0.931	0.034
VQGAN	10%	Orthogonal	19.73	0.803	0.141
VQGAN	10%	Random	21.37	0.810	0.137
VQGAN	10%	Farthest	12.96	0.732	0.222
VQGAN	25%	Closest	24.76	0.846	0.081
VQGAN	25%	Orthogonal	15.83	0.610	0.293
VQGAN	25%	Random	17.45	0.617	0.323
VQGAN	25%	Farthest	9.02	0.482	0.439
VQGAN	50%	Closest	22.43	0.741	0.141
VQGAN	50%	Orthogonal	13.12	0.384	0.477
VQGAN	50%	Random	14.84	0.431	0.557
VQGAN	50%	Farthest	6.39	0.247	0.667
VQGAN	75%	Closest	20.78	0.633	0.208
VQGAN	75%	Orthogonal	10.91	0.178	0.645
VQGAN	75%	Random	12.79	0.310	0.785
VQGAN	75%	Farthest	4.50	0.091	0.832

Table C.10: ImageNet-V2 decoder locality after token edits. Inside and outside are mean pixel RMSE relative to the clean reconstruction; leakage is the median per-image outside-to-inside ratio. The 99th percentile is the mean across images of each image’s outside-patch 99th percentile.

Tokenizer	Patch	Edit	Inside	Outside	Leakage	Outside P99
LlamaGen	10%	Closest	0.067	0.008	0.120	0.078
LlamaGen	10%	Orthogonal	0.210	0.023	0.109	0.194
LlamaGen	10%	Random	0.209	0.023	0.108	0.192
LlamaGen	10%	Farthest	0.334	0.029	0.088	0.253
LlamaGen	25%	Closest	0.067	0.013	0.190	0.102
LlamaGen	25%	Orthogonal	0.223	0.038	0.173	0.246
LlamaGen	25%	Random	0.221	0.038	0.172	0.245
LlamaGen	25%	Farthest	0.362	0.049	0.137	0.325
LlamaGen	50%	Closest	0.065	0.019	0.288	0.126
LlamaGen	50%	Orthogonal	0.229	0.060	0.265	0.298
LlamaGen	50%	Random	0.228	0.059	0.263	0.298
LlamaGen	50%	Farthest	0.384	0.077	0.206	0.402
LlamaGen	75%	Closest	0.064	0.027	0.411	0.151
LlamaGen	75%	Orthogonal	0.244	0.092	0.381	0.348
LlamaGen	75%	Random	0.241	0.090	0.378	0.348
LlamaGen	75%	Farthest	0.419	0.120	0.293	0.478
VQGAN	10%	Closest	0.069	0.011	0.162	0.106
VQGAN	10%	Orthogonal	0.227	0.028	0.124	0.250
VQGAN	10%	Random	0.187	0.029	0.154	0.228
VQGAN	10%	Farthest	0.593	0.052	0.086	0.543
VQGAN	25%	Closest	0.072	0.018	0.258	0.141
VQGAN	25%	Orthogonal	0.239	0.044	0.184	0.308
VQGAN	25%	Random	0.200	0.049	0.242	0.291
VQGAN	25%	Farthest	0.609	0.088	0.145	0.668
VQGAN	50%	Closest	0.071	0.027	0.385	0.174
VQGAN	50%	Orthogonal	0.247	0.066	0.267	0.364
VQGAN	50%	Random	0.209	0.073	0.353	0.341
VQGAN	50%	Farthest	0.613	0.138	0.229	0.744
VQGAN	75%	Closest	0.073	0.038	0.523	0.205
VQGAN	75%	Orthogonal	0.262	0.096	0.371	0.417
VQGAN	75%	Random	0.222	0.103	0.480	0.372
VQGAN	75%	Farthest	0.616	0.213	0.350	0.779

Table C.11: ImageNet-V2 variation across images for each token-edit condition. Columns are the saved per-condition standard deviations: patch PSNR in dB, patch LPIPS, and inside and outside pixel RMSE.

Tokenizer	Patch	Edit	PSNR SD	LPIPS SD	Inside SD	Outside SD
LlamaGen	10%	Closest	3.65	0.040	0.025	0.003
LlamaGen	10%	Orthogonal	2.20	0.079	0.055	0.006
LlamaGen	10%	Random	2.31	0.079	0.057	0.006
LlamaGen	10%	Farthest	2.11	0.100	0.091	0.008
LlamaGen	25%	Closest	2.98	0.040	0.022	0.005
LlamaGen	25%	Orthogonal	1.76	0.068	0.049	0.010
LlamaGen	25%	Random	1.83	0.067	0.050	0.010
LlamaGen	25%	Farthest	1.79	0.086	0.086	0.013
LlamaGen	50%	Closest	2.72	0.040	0.021	0.007
LlamaGen	50%	Orthogonal	1.57	0.061	0.046	0.015
LlamaGen	50%	Random	1.62	0.061	0.047	0.015
LlamaGen	50%	Farthest	1.62	0.080	0.083	0.019
LlamaGen	75%	Closest	2.58	0.040	0.020	0.011
LlamaGen	75%	Orthogonal	1.53	0.060	0.050	0.025
LlamaGen	75%	Random	1.57	0.060	0.050	0.025
LlamaGen	75%	Farthest	1.54	0.078	0.087	0.035
VQGAN	10%	Closest	3.79	0.052	0.027	0.005
VQGAN	10%	Orthogonal	2.05	0.087	0.056	0.009
VQGAN	10%	Random	2.61	0.103	0.057	0.010
VQGAN	10%	Farthest	1.33	0.089	0.103	0.013
VQGAN	25%	Closest	3.09	0.053	0.025	0.008
VQGAN	25%	Orthogonal	1.62	0.075	0.047	0.014
VQGAN	25%	Random	2.16	0.082	0.053	0.016
VQGAN	25%	Farthest	1.13	0.078	0.089	0.021
VQGAN	50%	Closest	2.80	0.054	0.023	0.011
VQGAN	50%	Orthogonal	1.41	0.068	0.043	0.019
VQGAN	50%	Random	1.98	0.074	0.052	0.022
VQGAN	50%	Farthest	1.07	0.075	0.084	0.028
VQGAN	75%	Closest	2.63	0.057	0.023	0.017
VQGAN	75%	Orthogonal	1.28	0.068	0.044	0.028
VQGAN	75%	Random	1.95	0.073	0.056	0.029
VQGAN	75%	Farthest	1.09	0.074	0.086	0.048

C.5. Distribution-shift artifact inventory

Table C.12: Encoded Chapter 8 artifacts. Every row has a 16×16 token grid and satisfies $\text{total tokens} = \text{images} \times 256$. Active codes count indices observed at least once on that dataset.

Tokenizer	Dataset	Images	Total tokens	Active codes
LlamaGen	BloodMNIST	17,092	4,375,552	16,124
LlamaGen	ImageNet	50,000	12,800,000	16,384
LlamaGen	ImageNet-Sketch	50,889	13,027,584	16,383
LlamaGen	ImageNet-V2	10,000	2,560,000	16,384
LlamaGen	ObjectNet	50,273	12,869,888	16,384
LlamaGen	OrganAMNIST	57,187	14,639,872	16,383
LlamaGen	RVL-CDIP	39,996	10,238,976	16,288
VQGAN	BloodMNIST	17,092	4,375,552	966
VQGAN	ImageNet	50,000	12,800,000	972
VQGAN	ImageNet-Sketch	50,889	13,027,584	972
VQGAN	ImageNet-V2	10,000	2,560,000	972
VQGAN	ObjectNet	50,273	12,869,888	972
VQGAN	OrganAMNIST	57,187	14,639,872	972
VQGAN	RVL-CDIP	39,996	10,238,976	971

C.6. Additional reported contrasts

Tokenizer	Outside mean	Outside P99	Outside P99.9
LlamaGen	0.050	0.328	0.554
VQGAN	0.090	0.673	0.854

Table C.13: Outside response for the 25% farthest-code edit on ImageNet. Each tail entry is the mean of the corresponding per-image outside-pixel percentile, in RGB RMSE units on $[0, 1]$. Values are from the archived decoder-locality summary.

Tokenizer	Dataset	All	Border	Interior
LlamaGen	ImageNet-V2	-0.682	-0.672	-0.686
LlamaGen	ImageNet-Sketch	-1.166	-2.212	-0.846
LlamaGen	ObjectNet	-0.298	-0.924	-0.106
LlamaGen	OrganAMNIST	-0.544	-0.613	-0.522
LlamaGen	BloodMNIST	-1.552	-2.007	-1.413
LlamaGen	RVL-CDIP	-1.668	-2.447	-1.429
VQGAN	ImageNet-V2	-0.039	-0.034	-0.041
VQGAN	ImageNet-Sketch	-0.904	-1.798	-0.630
VQGAN	ObjectNet	-0.131	-0.350	-0.064
VQGAN	OrganAMNIST	-0.249	-0.215	-0.259
VQGAN	BloodMNIST	-1.003	-1.259	-0.925
VQGAN	RVL-CDIP	-1.254	-2.025	-1.018

Table C.14: Mean positional-entropy change from the full ImageNet reference, in nats, from the archived positional summary. These are unequal-size descriptive estimates, not the matched estimates in Table 8.4. Border is the one-cell outer ring.

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List of Symbols

Symbol	Description
x	Input image.
$\hat{x}$	Reconstructed image.
x_σ	Image perturbed by Gaussian noise with standard deviation σ .
E, Q, D	Encoder, vector quantizer, and decoder.
z_e, z_q	Continuous encoder output and quantized latent representation.
z_0, z_σ	Clean and perturbed discrete token grids.
e_k	Embedding vector associated with codebook entry k .
K	Nominal codebook size.
h, w	Height and width of the token grid.
p_k	Empirical assignment probability of codebook entry k .
$H(p)$	Entropy of the empirical token distribution.
$P(p)$	Codebook perplexity, $\exp(H(p))$, also called effective vocabulary size.
n_k, N_t	Assignments to code k and total assignments in the evaluation set.
$A_{\text{raw}}, a_{\text{raw}}$	Number and fraction of codes observed at least once.
M_m, C_{99}	Probability mass of the top m codes, and number of codes needed to cover 99% of assignments.
k, t_i, F_i	Code index; token assignment at position i ; indicator that this assignment changes. A grid of indices is denoted k in Chapter 2 and z in the experimental chapters.
$r(R)$	Fraction of changed token assignments within a set of positions R .
σ	Standard deviation of an applied Gaussian perturbation.
M_P	Binary image-space mask for patch P .

I_P, O_P	Token positions inside and outside patch P .
$R_{\text{in}}, R_{\text{out}}$	Mean encoder-side token response inside and outside a perturbed patch.
λ_{enc}	Encoder leakage ratio, $R_{\text{out}}/(R_{\text{in}} + \varepsilon)$.
$d(i, P), R_d$	Manhattan distance from token position i to patch P , and flip rate at distance d .
$B_{\text{tok}}, B_{\text{px}}$	Edited token-space region and its corresponding image-space region.
$\Delta_{\text{in}}, \Delta_{\text{out}}$	Mean decoder-side image response inside and outside an edited region.
λ_{dec}	Decoder leakage ratio, $\Delta_{\text{out}}/(\Delta_{\text{in}} + \varepsilon)$.
T_m	Code replacement rule for edit mode m in Chapter 7.
Ret, R_P	Retained fraction of ImageNet-active codes, and shifted-to-reference perplexity ratio.
$H_{u,v}$	Code entropy in nats at grid position (u, v) across images.
ε	Leakage-ratio stabilizer, 10^{-8} ; distinct from the Gaussian noise variable ϵ .
N, m	Locally defined count and index: N counts scalar pixels or token positions in the relevant equation; m denotes top-mass cutoff, edit mode, or JSD mixture as specified locally.

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